Stratified Flow Distributions: Branch Mass, Occupancy, and Noise

Elvis Han Cui*
Department of Biostatistics, University of California, Los Angeles
Los Angeles, CA, USA

Kuntuo, an IQVIA Company
Hangzhou, China

Abstract

Probability mass, longitudinal occupancy, and transverse noise describe different changes in a distribution concentrated near branches. A decrease in local density can arise from any of them. Stratified flow distributions (SFDs) express these quantities separately through a mixture of declared branch laws, localized states, and residual components. We formulate a normalized finite-grid likelihood and characterize when a grouped branch mass is identified under fixed kernels, including cases with unidentified within-branch weights. Controlled two-arm experiments show that fixing a changed noise kernel produces a spurious mass contrast, while joint estimation separates the changes. Further experiments examine estimated directions, overlapping branches, noise misspecification, and parameter intervals. A training-fitted PBMC3k analysis evaluates posterior allocation as a feature-held-out prediction readout alongside generic mixture controls. The results clarify the role of kernel assumptions and geometric grouping in interpreting branch probabilities.

1 What Changed Near the Branch?

Consider two samples concentrated near the same branching structure. The second sample has a lower density peak near one branch. Three explanations are possible: the branch carries less probability, observations have moved along it, or their transverse spread has increased. These changes have different scientific meanings. For molecular configurations expressed in a common coordinate system, a change in the probability of a conformational region differs from a change in dispersion within that region. An analysis of branch-associated populations should distinguish the two.

The ambiguity is already visible on a straight branch. With tangent and normal coordinates (u, v), independent Gaussian observation noise gives the weighted component density

$$p_b(u, v) = \pi_b(h_b * \varphi_\epsilon)(u)\varphi_{\sigma_b}(v), \tag{1}$$

where h_b is a normalized longitudinal density and φ_σ is a centered Gaussian density with SD σ. Doubling σ_b halves the normal peak while preserving the integrated branch mass π_b. Halving π_b also halves the peak, but changes that mass. The longitudinal law h_b determines where observations accumulate. A geometric centerline or a peak-height comparison alone does not distinguish these mechanisms.

We study this question through stratified flow distributions (SFDs), whose components attach probability laws to declared atoms, branches, noisy tubes, and an ambient residual. The mixture construction supplies a common language for the probability mass of a component, its internal coordinate law, and its observation kernel. Its usefulness depends on which of these quantities the observations identify. In particular, a visually convincing branch does not by itself identify its mass when components overlap.

*AI-assisted tools were used in manuscript drafting and editing, code development, and numerical checks. The author is responsible for the scientific content and its verification.

This paper develops an estimand-oriented account of that distinction. A normalized finite-grid formulation separates branch mass from numerical occupancy weights. A fixed-kernel analysis identifies precisely which grouped masses are determined by the mixture density and gives their conditional information bound. Repeated experiments then quantify how kernel constraints affect mass recovery, how the result changes when two branch directions are estimated, and how allocation features compare with generic density features in a cell-level prediction task. The main empirical result is a specific failure of fixed kernels and its correction by joint estimation.

2 Related Work

Density models and geometric mixtures. Finite mixtures and KDE model ambient densities (McLachlan and Peel, 2000; Silverman, 1986). Normalizing flows, score-based diffusion and flow matching extend this toolkit through learned transformations or vector fields (Papamakarios et al., 2021; Song et al., 2021; Lipman et al., 2023); copulas separate marginal laws from dependence (Nelsen, 2006). These normalized families can supply SFD residuals or observation kernels. The SFD specification additionally names the component masses, internal coordinate laws and reference measures.

Generative topographic mapping (GTM) combines a smooth latent map, Gaussian observation kernels, EM fitting and posterior coordinates, and discusses mixtures of maps and learnable node probabilities (Bishop et al., 1998). Finite Gaussian SFDs share this kernel-mixture structure. Our analysis concerns the grouping of nodes into branch-mass targets and the effects of occupancy and noise constraints on those targets. A generic mixture supplied with the same grouping can estimate the same mass, which motivates matching the grouping in a comparison.

Geometry and trajectory models. Principal curves, Isomap and diffusion maps summarize low-dimensional structure (Hastie and Stuetzle, 1989; Tenenbaum et al., 2000; Coifman and Lafon, 2006); manifold estimation studies its recovery under noise (Genovese et al., 2012). Pseudotime and optimal-transport methods organize single-cell observations along developmental progressions (Trapnell et al., 2014; Schiebinger et al., 2019). These geometric summaries can motivate a declared skeleton. A probability law on that skeleton also specifies longitudinal occupancy, observation noise and residual mass. LAND uses a learned Riemannian metric to define locally adaptive normal laws (Arvanitidis et al., 2016), while Riemannian continuous flows define distributions through manifold-valued dynamics (Mathieu and Nickel, 2020). Their metric and volume conventions must be retained when such laws are used as mixture components.

Stratified spaces and inference. Tree space provides a geometric setting for observations with different combinatorial types (Billera et al., 2001). Sticky limit theorems on open books show how singular geometry can alter mean asymptotics (Hotz et al., 2013). In SFD, observations lie in an ambient space and the mixture allocates mass to embedded atoms, exact strata, noisy tubes and residuals. A common dominating measure makes their likelihood contributions comparable. The fixed-chart result below uses regular likelihood theory; its conditions distinguish that setting from moving singular supports and boundary models (van der Vaart, 1998).

3 The SFD Probability Law

Let $X \in \mathcal{X} \subseteq \mathbb{R}^d$ be the observed state vector. Finite, disjoint index sets $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{R}$ label atoms, branches and residual components; write $\mathcal{K} = \mathcal{A} \cup \mathcal{B} \cup \mathcal{R}$. A declared skeleton specifies atom locations ξ_a, branch maps $\gamma_b : \mathcal{S}_b \to \mathcal{X}$, and residual components with normalized densities q_r. Each compact coordinate domain $\mathcal{S}_b \subseteq \mathbb{R}^{m_b}$ has finite positive volume. Let h_b be a normalized coordinate density and $K_b(dx \mid s)$ a measurable probability kernel. Write η_b, ψ_b and ω_r for coordinate-law, kernel and residual parameters, respectively, and suppress their arguments when unambiguous. All component laws are supported on $\mathcal{X}$.

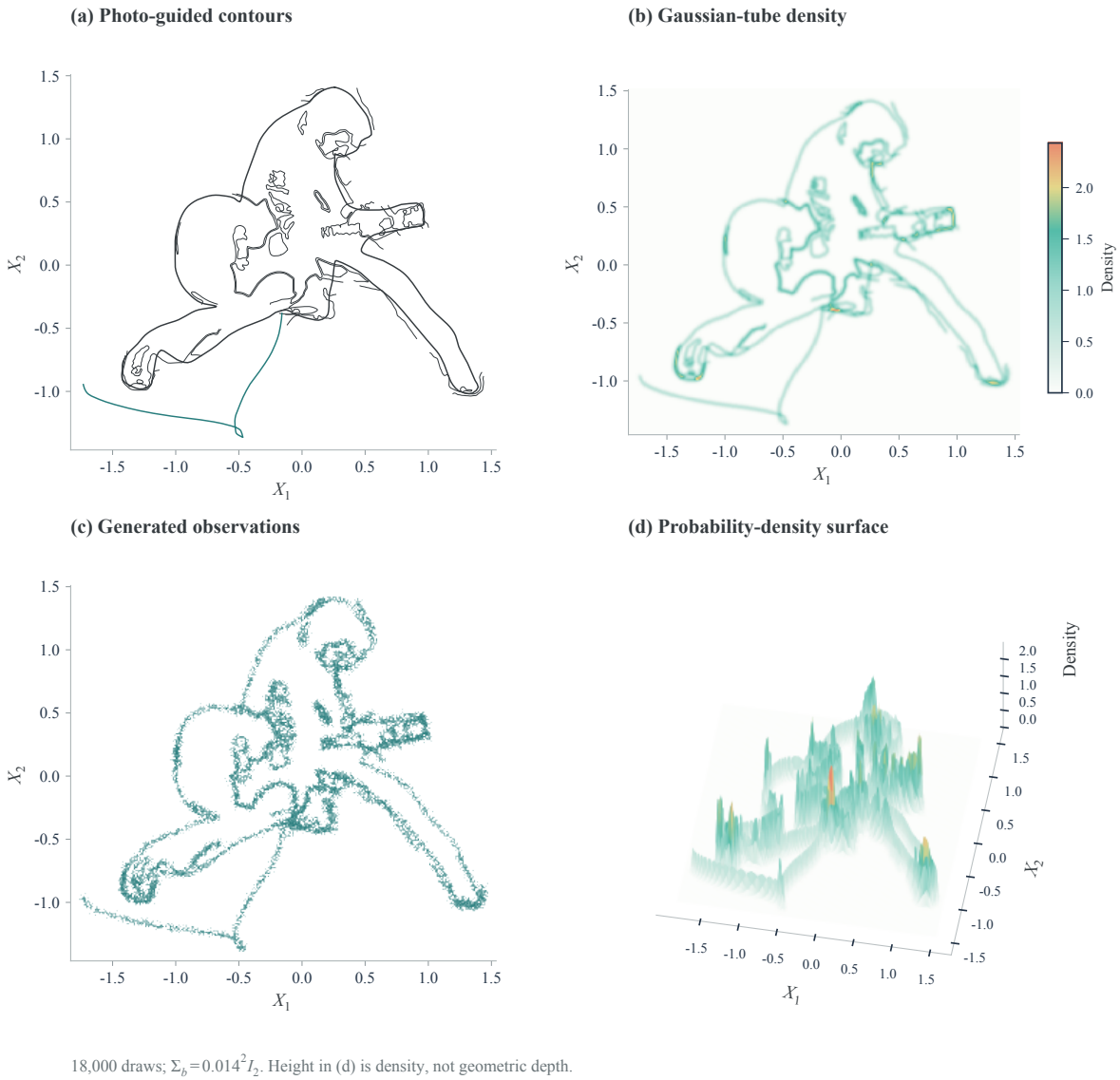


Figure 1: An Elvis pose as a 35-component SFD. (a) Photo-guided curves, with the cable in teal. (b) Gaussian-tube density with arclength weights and covariance $0.014^2 I_2$. (c) 18,000 generated observations. (d) The same density as a surface: height represents density, not depth.

Definition 1 (Stratified Flow Distribution (SFD))**.** An SFD, with Chinese alias 雄起流分布, is the probability law

$$P_\theta(dx) = \sum_{a\in\mathcal{A}} \pi_a \delta_{\xi_a}(dx) + \sum_{b\in\mathcal{B}} \pi_b \int_{\mathcal{S}_b} h_b(s;\eta_b) K_b(dx \mid s;\psi_b)\, ds + \sum_{r\in\mathcal{R}} \pi_r q_r(x;\omega_r)\, dx, \tag{2}$$

with $\pi_c \geq 0$ and $\sum_{c\in\mathcal{K}} \pi_c = 1$.

Each component integrates to one. Exact branches use $K_b(dx \mid s) = \delta_{\gamma_b(s)}(dx)$; full-rank Gaussian tubes use $K_b(dx \mid s) = \mathcal{N}(dx; \gamma_b(s), \Sigma_b(s))$ with $\Sigma_b(s) \in \mathbb{R}^{d\times d}$ positive definite. A localized Gaussian blob is a continuous component, distinct from a Dirac atom.

For a fixed finite skeleton, a common dominating measure ν can include Lebesgue measure, atom measures, and coordinate-pushforward measures of exact branches. Densities in $p_\theta = dP_\theta/d\nu$ must all use that common reference. If every component is regularized, Lebesgue measure suffices. The experiments below use full-rank densities in common observed coordinates; the mixed-measure formulation also covers exact strata. A moving singular support requires a separate common-reference argument.

Proposition 1 (Master expectation identity)**.** *For measurable g with $\mathbb{E}_\theta|g(X)| < \infty$,*

$$\begin{aligned}\mathbb{E}_\theta g(X) = \sum_a \pi_a g(\xi_a) + \sum_b \pi_b \int_{\mathcal{S}_b} h_b(s) \int g(x) K_b(dx \mid s)\, ds \\ + \sum_r \pi_r \int g(x) q_r(x)\, dx.\end{aligned} \tag{3}$$

Condition on the component and then its coordinate to obtain the identity; Tonelli–Fubini gives its integrable signed form. Appendix D gives normalization, moments, mixed densities, marginals and conditionals.

Three distinct summaries. With component label Z and branch coordinate S, the mass and longitudinal mean are

$$\pi_b = \mathbb{P}(Z = b), \qquad \bar{s}_b = \mathbb{E}(S \mid Z = b) = \int_{\mathcal{S}_b} s h_b(s)\, ds.$$

The transverse part of the observation kernel describes spread around the branch. These are latent-component quantities: overlapping tubes and residuals also contribute probability to any observed neighborhood of a curve. Consequently, that neighborhood probability generally differs from π_b. Figure 1 turns selected image-plane curves into a concrete SFD by declaring their masses, coordinate laws and kernels.

Local geometry and coordinate conventions. For a regular curve, let $t_b(s) \in \mathbb{R}^d$ be its unit tangent and $N_b(s) \in \mathbb{R}^{d\times(d-1)}$ an orthonormal normal frame. Independent centered Gaussian perturbations along and normal to the curve give

$$\begin{aligned}X &= \gamma_b(S) + t_b(S)\varepsilon_{\parallel,b} + N_b(S)\varepsilon_{\perp,b}, \\ \Sigma_b(s) &= \sigma^2_{\parallel,b}(s) t_b(s) t_b(s)^\top + N_b(s)\Sigma_{\perp,b}(s) N_b(s)^\top.\end{aligned} \tag{4}$$

For a one-to-one regular C^1 exact branch with image Γ_b, the conditional law $P_b = (\gamma_b)_\#\{h_b(s)ds\}$ has arc-length density

$$\frac{dP_b}{d(\mathcal{H}^1|_{\Gamma_b})}\{\gamma_b(s)\} = \frac{h_b(s)}{\|\dot\gamma_b(s)\|}. \tag{5}$$

Transporting the coordinate density and kernel under an increasing smooth reparameterization preserves the branch law and mass. Coordinate means require a common chart; normalized arc length removes coordinate warping when origin, orientation, and branch correspondence are shared. Appendix B gives the transformations and distinguishes latent skeleton connectivity from the support of noisy observations.

Allocation and conditioning. For regularized component densities g_c,

$$\mathbb{P}(Z = c \mid X = x) = \frac{\pi_c g_c(x)}{\sum_{c'} \pi_{c'} g_{c'}(x)}. \tag{6}$$

For an event A of positive probability, its component masses become $\pi_c P_c(A)/\sum_{c'} \pi_{c'} P_{c'}(A)$. Marginals, event probabilities, and conditional simulation follow from the same mixture law. The fitted occupancy describes a static distribution; a model for time-ordered observations is needed to estimate transition rates or probability flux.

Proposition 2 (Single-trajectory tube special case)**.** *Setting $\mathcal{A} = \mathcal{R} = \emptyset$, $\mathcal{B} = \{1\}$, $\pi_1 = 1$ and $K_1(dx \mid s) = \mathcal{N}(dx; \gamma_1(s), \Sigma_1(s))$ with positive-definite $\Sigma_1(s)$ gives exactly*

$$p_{\text{tube}}(x) = \int_{\mathcal{S}_1} h_1(s)\phi_d\{x; \gamma_1(s), \Sigma_1(s)\}\, ds,$$

where ϕ_d denotes the d-dimensional Gaussian density.

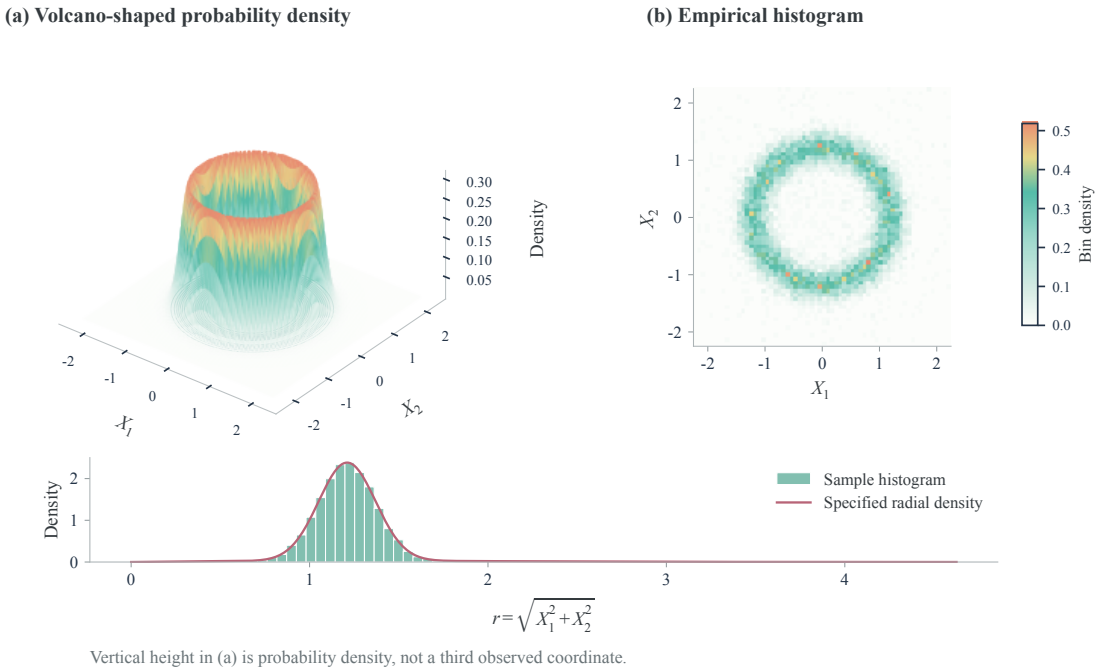


Figure 2: A circular Gaussian tube produces a volcano-shaped density. (a) The specified density. (b) A bivariate histogram of 16,000 draws, normalized by the full sample size. Bottom: sampled radii and their density $2\pi rp(r)$.

This follows by substitution in Definition 1. Appendix D.3 gives the other layer restrictions.

A circular branch and a volcano-shaped density. Take $\gamma(s) = 1.2(\cos 2\pi s, \sin 2\pi s)$, uniform $S \in [0, 1]$, kernel covariance $0.16^2 I_2$ and branch mass 0.94. A centered Gaussian residual has mass 0.06 and covariance $1.25^2 I_2$. The resulting Lebesgue density concentrates around the ring while its centerline carries zero observation probability. Figure 2 displays the density, an empirical histogram and the radial law. The construction separates a branch's geometric image from its probability mass and observation width.

Projected geometry and reference samples. Figure 3 uses a curved section of the Fermat quintic, a Calabi–Yau hypersurface (Hendi et al., 2024). The section is projected to $\mathbb{R}^3$ and approximated by 325 Gaussian tubes with equal masses and covariance $0.035^2 I_3$. The projection and coordinate-grid law are given in Appendix B.1. This example has three observed coordinates, unlike the density heights in Figures 1d and 2a.

Figure 4 complements these declared-geometry examples with empirical views of Gaussian, skew-mixture and banana-transform reference samples. Joint and marginal histograms expose different aspects of the same sample shape. Each display uses 500 stored training observations; the fitted-method comparisons begin in Section 6. Appendix C records the construction details for Figures 1–4.

Density surfaces. Figure 5 shows four specified bivariate laws, including a Doraemon-style face. These are density surfaces over two coordinates, not three-dimensional observation models. Appendix C records the display transformation.

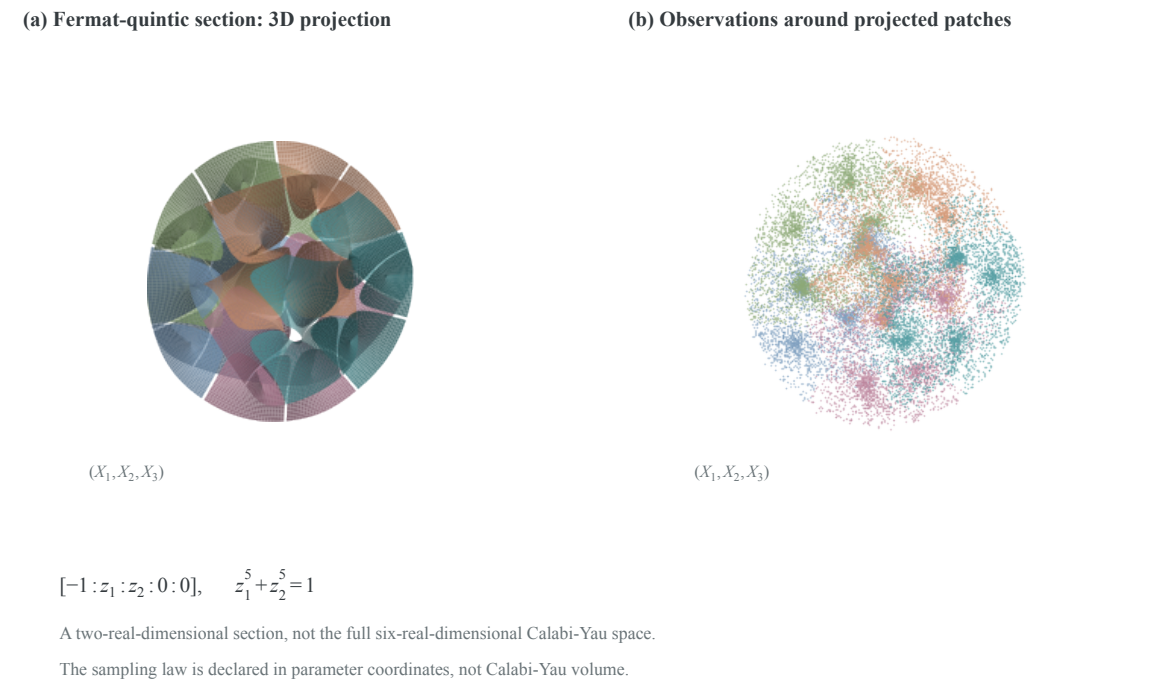


Figure 3: A projected Fermat-quintic section and an SFD around it. Left: the geometric patches, colored by root index. Right: 18,000 observations from 325 Gaussian tubes. Both panels use three spatial coordinates.

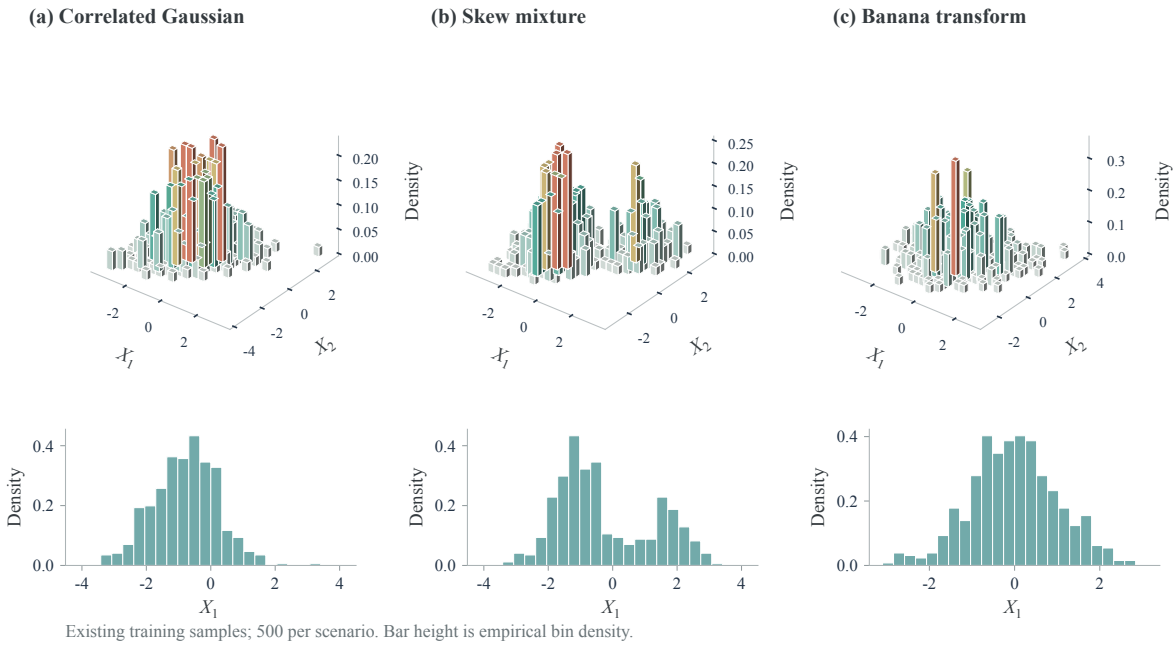


Figure 4: Reference-sample histograms for a correlated Gaussian, skew mixture and banana transform. Top: 24×24 bin densities in the first two coordinates. Bottom: first-coordinate marginals. Each scenario uses all 500 training draws.

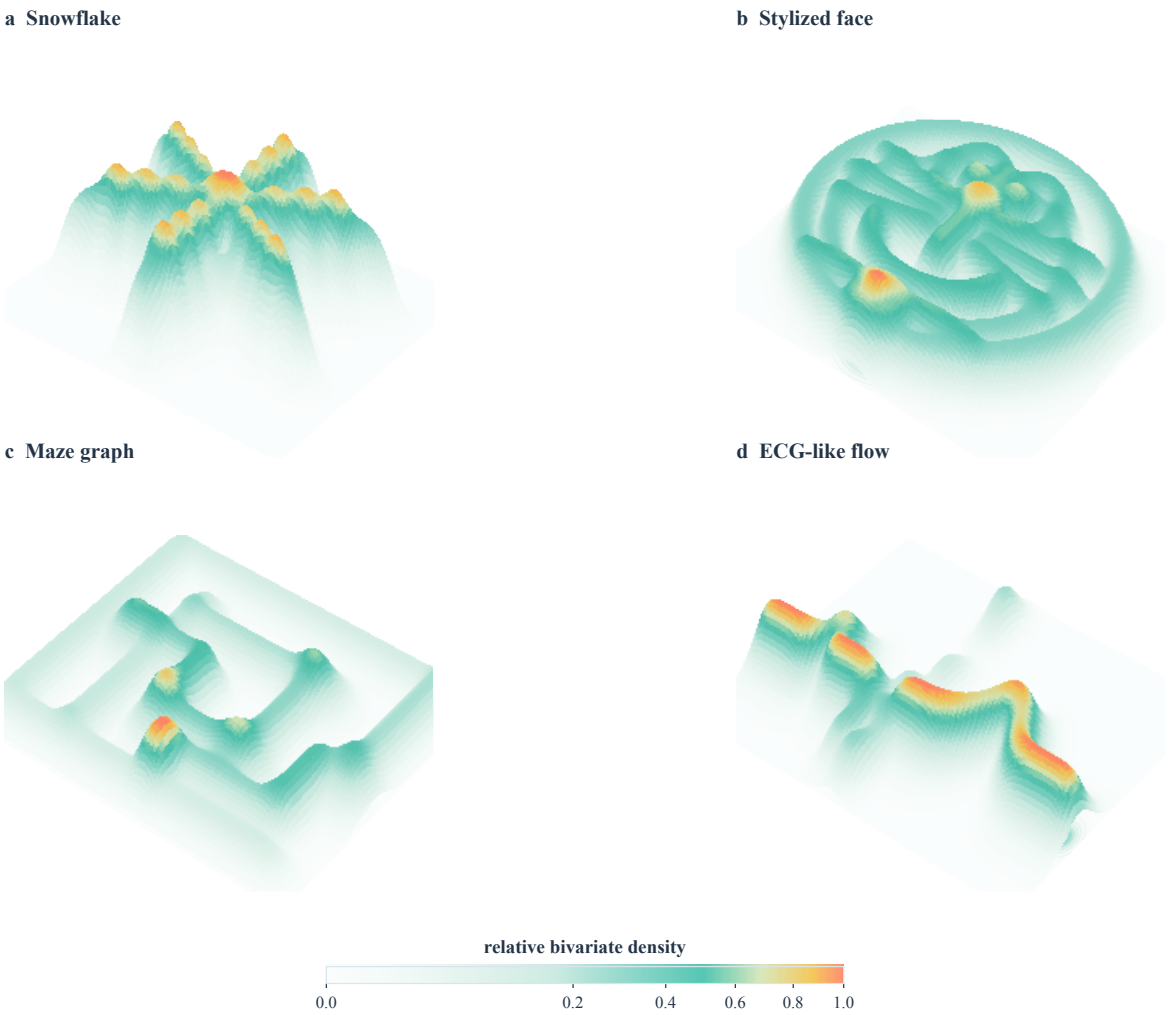


Figure 5: Specified snowflake, Doraemon-style face, maze and ECG-like densities. Height and color use a separate nonlinear relative scale in each panel, revealing shape rather than comparing absolute density. Curves contribute Gaussian tubes; localized peaks are Gaussian blobs.

4 Normalized Finite-Skeleton Estimation

Positive quadrature weights w_{bm} and coordinate nodes s_{bm} give

$$C_b(\eta_b) = \sum_m w_{bm} h_b(s_{bm}; \eta_b), \qquad \kappa_{bm}(\eta_b) = \frac{w_{bm} h_b(s_{bm}; \eta_b)}{C_b(\eta_b)}, \qquad g_b^{\mathrm{grid}}(x) = \sum_m \kappa_{bm} k_b(x \mid s_{bm}). \tag{7}$$

For $C_b > 0$, the node probabilities sum to one and each finite-grid branch is normalized. Normalization is essential when h_b changes: otherwise its quadrature error can enter the apparent branch mass. Accuracy relative to the continuous integral remains a grid-resolution question.

The declared objective is

$$F_n(\theta) = \sum_i \log p_\theta^{\mathrm{grid}}(x_i) - \mathrm{Pen}(\theta). \tag{8}$$

Node responsibilities are $r_{ibm} = \pi_b \kappa_{bm} k_b(x_i \mid s_{bm}) / p_\theta^{\mathrm{grid}}(x_i)$. Writing $n_b^{\mathrm{eff}} = \sum_{i,m} r_{ibm}$, the occupancy block of the EM surrogate contains

$$\sum_{i,m} r_{ibm} \log h_b(s_{bm}; \eta_b) - n_b^{\mathrm{eff}} \log C_b(\eta_b). \tag{9}$$

Its second term must remain in the optimization. A Dirichlet penalty on outer masses yields updates proportional to $n_c^{\mathrm{eff}} + \alpha_c - 1$ when $\alpha_c > 1$; penalties on other parameters remain in their respective blocks.

Algorithm 1 Objective-checked finite-grid estimation

Require: Observations, declared branches and component families, positive quadrature rule, penalty, feasible starts, and stopping tolerances.

1: **for** each start **do**
2: Normalize node probabilities κ_{bm} and evaluate F_n.
3: **repeat**
4: Compute responsibilities for all nodes and residual components.
5: Update masses and propose occupancy, geometry, and noise blocks.
6: Retain the normalizer in (9) and all penalties.
7: Accept feasible block steps only if they do not decrease F_n.
8: **until** objective and stationarity tolerances hold, or the iteration limit is reached
9: Record the objective, stopping status, and any active constraints.
10: **end for**
11: Select the feasible fit with largest recorded objective; retain all statuses.

Proposition 3 (Fixed-quadrature GEM monotonicity)**.** *Fix the normalized quadrature law, latent label set, parameter space and penalty. Suppose the observed densities are finite and positive at the old and proposed iterates. An update that does not decrease the penalized expected complete-data objective cannot decrease F_n.*

The proof is the EM lower-bound argument (Dempster et al., 1977; Wu, 1983), given in Appendix F. Algorithm 1 states the objective checks for approximate block updates. After covariance factorization, a responsibility pass costs $O(nMd^2)$ for M dense-covariance nodes, or $O(nMd)$ for isotropic kernels. This per-pass calculation leaves the number of iterations and nonconvex optimization behavior to the selected parameterization.

With fixed kernels, weight estimation is concave. A positive pseudocount penalty can select a unique maximizer even when some likelihood weights are unidentified. The next section distinguishes optimization uniqueness from identification. The two-arm experiments use closed-form longitudinal convolution and direct likelihood optimization. The PBMC analysis uses a specified MST width/residual heuristic; its small-mass-change stopping rule is reported separately from objective-checked GEM.

5 Identification and Fixed-Stratum Inference

5.1 Identification of a Grouped Mass

Fix normalized kernels $f_1, \dots, f_K$ relative to ν, their grouping, and an interior simplex weight $w \in \mathbb{R}^K$. Write $p_w = \sum_j w_j f_j$ and $T_\zeta(w) = \zeta^\top w$. For a branch mass, ζ indicates the nodes assigned to that branch. Define the likelihood-null directions

$$\mathcal{V}_0 = \{v \in \mathbb{R}^K : \mathbf{1}^\top v = 0, \textstyle\sum_j v_j f_j = 0 \ \nu\text{-a.e.}\}. \tag{10}$$

Proposition 4 (Fixed-kernel grouped target)**.** *The target T_ζ is identified on the interior simplex exactly when $\zeta \perp \mathcal{V}_0$. Eliminate w_K and set, for $j < K$,*

$$U_j(x) = \frac{f_j(x) - f_K(x)}{p_w(x)}, \qquad I_w = \mathbb{E}_w\{U(X)U(X)^\top\}, \qquad \beta_{\zeta,j} = \zeta_j - \zeta_K, \tag{11}$$

with $U = 0$ where $p_w = 0$. Identification is equivalent to $\beta_\zeta \in \mathrm{range}(I_w)$. Every centered square-integrable influence function ϕ satisfying $\mathbb{E}_w(\phi U) = \beta_\zeta$ obeys

$$\mathbb{E}_w(\phi^2) \geq \beta_\zeta^\top I_w^+ \beta_\zeta, \tag{12}$$

where I_w^+ is the Moore-Penrose pseudoinverse.

Equal mixture densities have weight difference in $\mathcal{V}_0$. Orthogonality therefore gives sufficiency, and a small interior perturbation along a null direction gives necessity. The null space of I_w describes the same density-preserving directions; projection onto the score span gives the bound. Appendix A supplies the proof. The result applies standard mixture-identification and information arguments (McLachlan and Peel, 2000; van der Vaart, 1998) to the declared grouped target.

If $f_1 = f_2$ within one branch, their separate weights are unidentified but their sum can remain identified when the branch separates from the other kernels. Assigning those duplicate kernels to different branches instead destroys identification of the separate masses. Refining a within-branch grid and increasing cross-branch overlap can therefore have different consequences for the same target.

This information bound conditions on fixed kernels. Estimating noise, geometry, residuals, or the grouping introduces additional nuisance directions. Prior curvature can select among observationally equivalent weights but cannot identify them from the likelihood. For fitted six- and eight-parameter two-arm models, Section 6.3 instead evaluates intervals using the complete estimated parameter chart.

5.2 Regular Fixed-Stratum Likelihood Theory

Joint estimation follows regular likelihood theory on an identifiable fixed-stratum chart (van der Vaart, 1998).

Theorem 1 (Regular finite-SFD likelihood theory)**.** *Let $X_i \overset{\text{iid}}{\sim} P_{\theta_0}$ in a finite SFD chart satisfying conditions* (R1)–(R5) *of Appendix E.1. Put $U_0 = \nabla \log p_{\theta_0}$ and $I_0 = \mathbb{E}_{\theta_0}(U_0 U_0^\top) \succ 0$. The consistent interior likelihood maximizer $\hat{\theta}_n$ satisfies*

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = I_0^{-1} n^{-1/2} \sum_i U_0(X_i) + o_p(1) \ \rightsquigarrow \ \mathcal{N}(0, I_0^{-1}). \tag{13}$$

For every differentiable fixed-stratum target ψ,

$$\sqrt{n}\{\psi(\hat{\theta}_n) - \psi(\theta_0)\} \ \rightsquigarrow \ \mathcal{N}\big(0, \dot{\psi}(\theta_0) I_0^{-1} \dot{\psi}(\theta_0)^\top\big)\,. \tag{14}$$

The proof and regular contrast tests are in Appendix E; Appendix F treats numerical-likelihood transfer.

6 Experiments

Four studies assess weight estimation on a supplied skeleton, target recovery, sensitivity to geometry and noise, and allocation as a prediction feature. Each uses 25 repetitions. Simulation repetitions are independent; PBMC splits overlap within one cell collection. Numbers in parentheses are Monte Carlo SEs of displayed means. Generating labels are used for evaluation, not fitting. Appendix H specifies optimizers, controls, preprocessing, and stopping rules; all prespecified density and marker controls appear in Appendix J.

6.1 Known-Skeleton Weight Estimation

Five specified laws use snowflake, stylized-face, maze, ECG-like, and torus-knot geometries. Each repetition provides 1000 training observations and 1000 independent test observations. EM estimates the component masses with the generating geometry, coordinate laws and noise kernels fixed; each tube integral uses 64 coordinate nodes. Figure 6 shows test log density, component-mass RMSE and component-label accuracy. The oracle evaluates the generating law, rather than an estimated model. This diagnostic isolates mass estimation from geometry and noise estimation; the next two studies examine the latter distinction. Appendix K gives classical-distribution and copula reference controls separately.



Figure 6: Weight-only SFD fitting on five supplied skeletons. Top: test log density for generic controls, fitted SFD and the oracle. Bottom: fitted component-mass RMSE and component-label accuracy. Bars show one Monte Carlo SE over 25 repetitions. The generating geometry and kernels are fixed; truth labels are used only for evaluation. GMM-16 and GMM-32 results are retained in the accompanying summary data.

6.2 When a Width Change Becomes a Mass Change

Two unit-length arms meet at angles $\pm\pi/6$. Each branch coordinate is $\mathcal{N}(\mu_b^{\mathrm{loc}}, 0.16^2)$ truncated to $[0,1]$. Observation noise has tangent SD 0.025 and normal SD σ_b. The residual is $\mathcal{N}((0.4,0),0.55^2 I_2)$. The reference

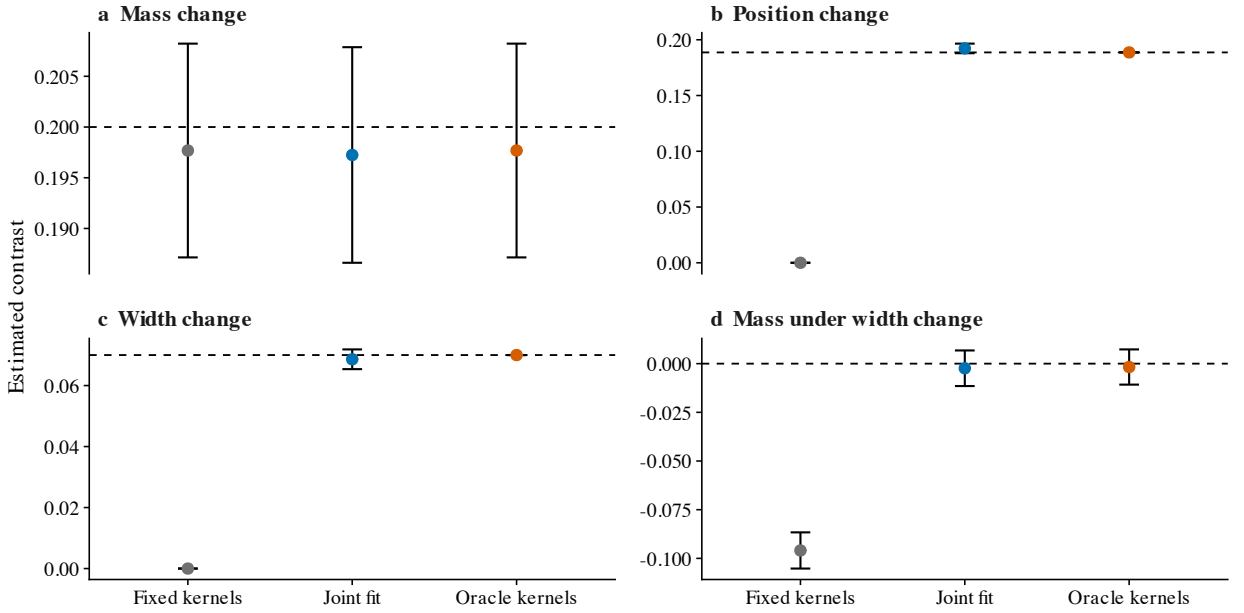


Figure 7: Two-arm target recovery. Panels a–c show mass, coordinate-mean and width contrasts; d shows the mass contrast under a width-only change. Dashed lines mark truth. Points and bars give means and ± 1.96 Monte Carlo SEs over 25 repetitions. The two kernel ablations estimate masses only.

has masses $(0.4, 0.4, 0.2)$, locations $(0.5, 0.5)$, and widths $(0.07, 0.07)$. Three alternatives change only masses to $(0.6, 0.2, 0.2)$, the upper location to 0.7, or the upper width to 0.14. Each group supplies 1000 training and 2000 independent test observations per repetition. The location parameter and truncated coordinate mean are distinct; the position alternative changes the latter by about 0.189.

The joint fit estimates two mass logits, two occupancy locations, and two normal widths, with origin, directions, coordinate scale, tangent noise, and residual law supplied. Two mass-only ablations fix either the reference kernels or the true group-specific kernels. This comparison isolates the consequence of fixing a kernel that changes between groups.

The joint estimates are 0.197 (0.005) for the mass contrast, 0.192 (0.002) for position, and 0.0686 (0.0017) for width, against generating contrasts 0.200, 0.189, and 0.0700. Under the width-only alternative the true mass contrast is zero, yet fixed reference kernels produce -0.0959 (0.0047). Joint estimation produces -0.0024 (0.0047) (Figure 7). The fixed-width model explains some increased transverse spread by lowering the fitted branch mass. All 300 selected SFD fits meet their optimizer stopping rule without parameter-bound contacts.

Gaussian, GMM-4, GMM-8, and KDE controls are scored on test log density and the observable probability $\mathbb{P}(X_2 > 0)$. Their components are not assigned branch labels. These controls assess prediction, whereas the joint-versus- fixed comparison assesses kernel constraints on a specified latent target. Every run, including GMM iteration-limit warnings, is retained in the supplementary score tables.

6.3 Estimated Directions and Misspecified Noise

The four group laws are repeated with Gaussian arms at $\pm 30°$, overlapping arms at $\pm 15°$, and $\pm 30°$ arms with transverse Student-t_5 noise scaled to the same SD. Each condition has the same sample sizes and 25 repetitions. Fits use correct directions, rotate both by $5°$, or estimate two angles from training coordinates. The learned-direction fit has eight parameters and retains the supplied origin, two-arm family, coordinate and tangent scales, and residual law.

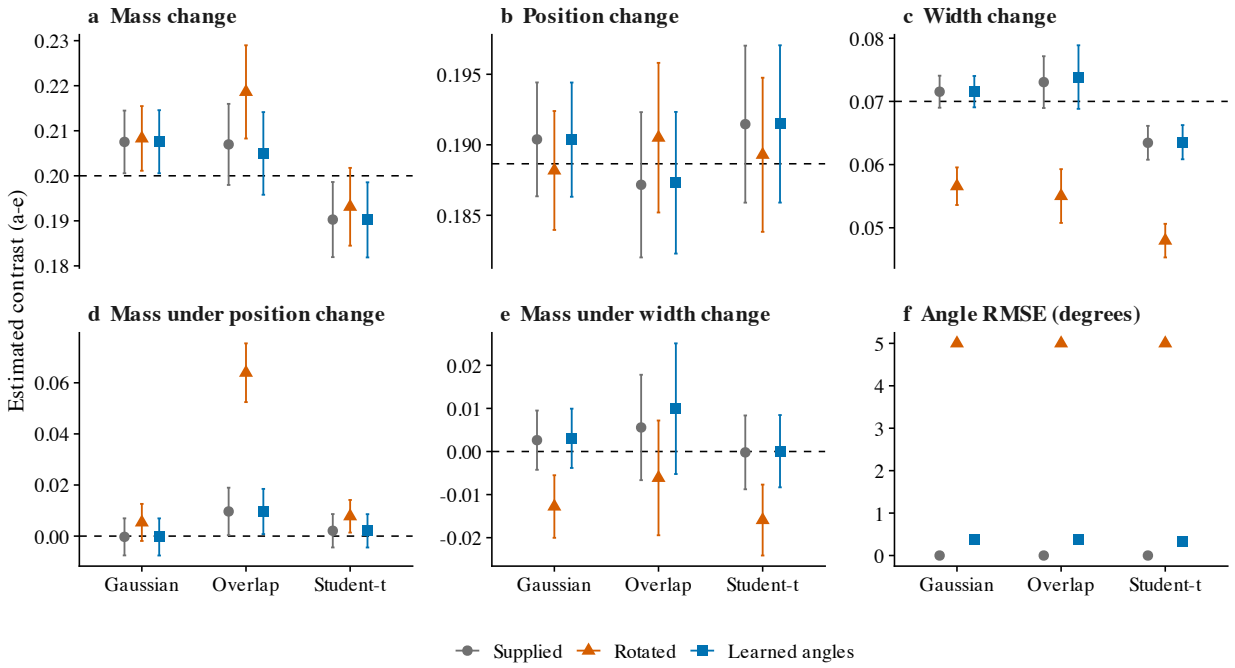


Figure 8: Geometry and noise sensitivity under supplied, 5°-rotated or learned directions. Panels a–e show target contrasts and mass cross-effects; f shows angle RMSE in degrees. Dashed lines mark truth; bars give ±1.96 Monte Carlo SEs over 25 repetitions. All fitted tubes are Gaussian, including those fitted to Student-t_5 noise.

The learned-angle reference RMSEs are 0.372 (0.034), 0.385 (0.045), and 0.332 (0.042) degrees. Estimated width contrasts are 0.0715 (0.0013), 0.0739 (0.0026), and 0.0635 (0.0014), against 0.0700. Under heavy-tailed noise, learning directions leaves a systematic width discrepancy under the Gaussian fitted family. Figure 8 also shows the effects of overlap and direction error on mass contrasts. All 900 selected SFD fits meet their stopping rule without boundary contacts; selected GMM fits converge under the expanded budget.

Parameter intervals. A post-hoc calculation uses each saved fit's complete six- or eight-parameter information matrix, including angles when estimated. With learned directions and Gaussian noise, nominal 95% Wald intervals cover generating mass, position, and width contrasts in 24/25, 24/25, and 25/25 repetitions. For rotated directions with Student-t_5 noise, width-contrast coverage is 3/25 with information covariance and 4/25 with sandwich covariance. The latter adjusts estimated variability, not misspecification of the fitted center. These parameter intervals are distinct from the Monte Carlo bars in Figure 8. Appendix I reports the full calculation, numerical checks, and coverage counts. Each count has only 25 repetitions, so even 25/25 provides limited precision about coverage.

6.4 Allocation as a Feature-Held-Out Readout

The public PBMC3k collection (10x Genomics, 2016) supplies an internal prediction study. In each of 25 splits, 2160 cells train the representation and 540 are held out. Eighteen predeclared marker genes are excluded from predictor normalization and selection. Training cells alone determine gene filtering, the 1000 selected nonmarker genes, scaling, clipping, and two-component PCA. Test cells use those transforms. Density scores refer to this two-dimensional representation, not the gene-count space.

An MST-skeleton tube with Gaussian residual is compared with Gaussian, GMM, and KDE densities. Mean test log density is −5.770 (0.018) for SFD, −5.367 (0.009) for GMM-8, and −5.421 (0.009) for KDE; generic

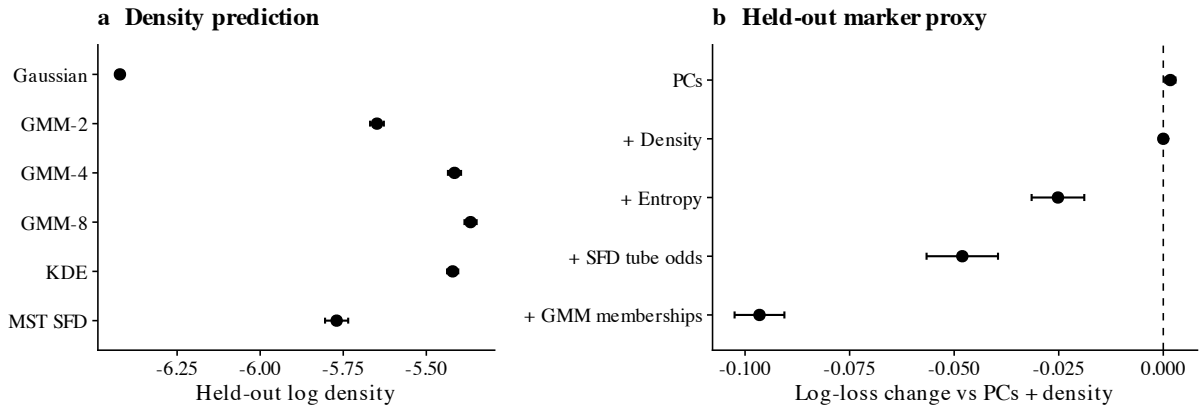


Figure 9: PBMC readouts on training-fitted PCs. a: test log density. b: paired log-loss changes for the held-out B-marker proxy, relative to PCs plus KDE density. SFD adds one tube-odds feature; GMM adds seven memberships. Bars give ±1.96 split Monte Carlo SEs over 25 overlapping cell splits, not donor-level uncertainty.

controls give the stronger density predictions here (Figure 9). The allocation question is whether a posterior tube feature adds information beyond the two PCs and KDE log density.

The binary outcome is a predeclared B-marker proxy computed from the held-out marker panel. Logistic readouts share the inner validation cells and penalty grid. Adding scalar SFD tube log odds changes mean test log loss from 0.2509 (0.0033) to 0.2029 (0.0056), a paired difference of −0.0481 (0.0044). GMM posterior entropy gives 0.2257 (0.0041); seven GMM membership coordinates give 0.1544 (0.0024). Allocation features therefore improve this readout beyond density features, with the larger generic-mixture representation performing better. These unequal- dimension readouts predict a marker proxy within one collection; they do not estimate independent-donor uncertainty.

7 Discussion

SFD separates branch probability, longitudinal occupancy, and observation noise within one declared probability law. The two-arm experiments show why this separation matters: a fixed-width constraint can turn increased spread into an apparent mass decrease, and estimating both blocks corrects that error under the specified model. The identification analysis explains why grouped masses can remain meaningful when numerical subdivisions are unidentified, and where overlap prevents that interpretation.

The interpretation conditions on a declared skeleton and kernel family. The experiments supply a two-arm family or a particular MST representation; arbitrary topology recovery is a separate problem. PBMC shows allocation features within a training-fitted representation, with stronger generic controls retained. Scientific state interpretation requires independently defined states and sampling units. A prospective time-resolved model would add a latent stochastic process and temporal observations (Appendix B).

Code and Data Availability

The software, experiment scripts, saved per-repetition results, and manuscript sources are available at

`https://github.com/ElvisCuiHan/stratified-flow-distributions`

The repository includes instructions for checking numerical summaries, regenerating figures, and rerunning experiments. PBMC3k raw data are available from 10x Genomics (10x Genomics, 2016); preprocessing and analysis settings are given in Appendix H.

Figure 1 reconstructs a pose from an author-supplied Elvis Presley photograph; the photograph itself is not reproduced. Figures 5b and 10 use a Doraemon-inspired synthetic geometry.

License

The author-authored text and original scientific material are licensed under Creative Commons Attribution 4.0 International. This license excludes third-party rights in the Elvis-derived illustration and Doraemon-inspired examples. Their inclusion does not grant permission to reuse those elements. Document styles retain their own license notices in the compilation archive. The original software is released separately under the MIT license.

References

10x Genomics (2016). 3k PBMCs from a healthy donor. *10x Genomics datasets*. `https://www.10xgenomics.com/datasets/3-k-pbm-cs-from-a-healthy-donor-1-standard-1-1-0`. Accessed 25 June 2026.

Arvanitidis, G., Hansen, L. K., and Hauberg, S. (2016). A locally adaptive normal distribution. *Advances in Neural Information Processing Systems*, 29.

Billera, L. J., Holmes, S. P., and Vogtmann, K. (2001). Geometry of the space of phylogenetic trees. *Advances in Applied Mathematics*, 27(4), 733–767.

Bishop, C. M., Svensen, M., and Williams, C. K. I. (1998). GTM: The generative topographic mapping. *Neural Computation*, 10(1), 215–234.

Boucheron, S., Lugosi, G., and Massart, P. (2013). *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press.

Coifman, R. R. and Lafon, S. (2006). Diffusion maps. *Applied and Computational Harmonic Analysis*, 21(1), 5–30.

Dempster, A. P., Laird, N. M., and Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B*, 39(1), 1–38.

Genovese, C. R., Perone-Pacifico, M., Verdinelli, I., and Wasserman, L. (2012). Manifold estimation and singular deconvolution under Hausdorff loss. *The Annals of Statistics*, 40(2), 941–963.

Hastie, T. and Stuetzle, W. (1989). Principal curves. *Journal of the American Statistical Association*, 84(406), 502–516.

Hendi, Y., Larfors, M., and Walden, M. (2024). Learning group invariant Calabi–Yau metrics by fundamental domain projections. *arXiv:2407.06914v2*. `https://arxiv.org/abs/2407.06914`.

Hotz, T., Huckemann, S., Le, H., Marron, J. S., Mattingly, J. C., Miller, E., Nolen, J., Owen, M., Patrangenaru, V., and Skwerer, S. (2013). Sticky central limit theorems on open books. *The Annals of Applied Probability*, 23(6), 2238–2258.

Kostrykin, V., Potthoff, J., and Schrader, R. (2012). Brownian motions on metric graphs. *arXiv:1102.4937v3*. `https://arxiv.org/abs/1102.4937`.

Lipman, Y., Chen, R. T. Q., Ben-Hamu, H., Nickel, M., and Le, M. (2023). Flow matching for generative modeling. *International Conference on Learning Representations*.

Mathieu, E. and Nickel, M. (2020). Riemannian continuous normalizing flows. *Advances in Neural Information Processing Systems*, 33.

McLachlan, G. and Peel, D. (2000). *Finite Mixture Models*. Wiley.

Nelsen, R. B. (2006). *An Introduction to Copulas*. Second edition. Springer.

Papamakarios, G., Nalisnick, E., Rezende, D. J., Mohamed, S., and Lakshminarayanan, B. (2021). Normalizing flows for probabilistic modeling and inference. *Journal of Machine Learning Research*, 22(57), 1–64.

Schiebinger, G., Shu, J., Tabaka, M., Cleary, B., Subramanian, V., Solomon, A., Gould, J., Liu, S., Lin, S., Berube, P., Lee, L., Chen, J., Brumbaugh, J., Rigollet, P., Hochedlinger, K., Jaenisch, R., Regev, A., and Lander, E. S. (2019). Optimal-transport analysis of single-cell gene expression identifies developmental trajectories in reprogramming. *Cell*, 176(4), 928–943.e22.

Silverman, B. W. (1986). *Density Estimation for Statistics and Data Analysis*. Chapman and Hall.

Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., and Poole, B. (2021). Score-based generative modeling through stochastic differential equations. *International Conference on Learning Representations*.

Tenenbaum, J. B., de Silva, V., and Langford, J. C. (2000). A global geometric framework for nonlinear dimensionality reduction. *Science*, 290(5500), 2319–2323.

Trapnell, C., Cacchiarelli, D., Grimsby, J., Pokharel, P., Li, S., Morse, M., Lennon, N. J., Livak, K. J., Mikkelsen, T. S., and Rinn, J. L. (2014). The dynamics and regulators of cell fate decisions are revealed by pseudotemporal ordering of single cells. *Nature Biotechnology*, 32(4), 381–386.

van der Vaart, A. W. (1998). *Asymptotic Statistics*. Cambridge University Press.

Wu, C. F. J. (1983). On the convergence properties of the EM algorithm. *The Annals of Statistics*, 11(1), 95–103.

A Fixed-Kernel Identification and Information

The result in Proposition 4 holds for arbitrary normalized nonnegative kernels under a fixed reference measure, including mixed measures. It is conditional on their scientific grouping and on an interior weight.

If $p_w = p_{w'}$, then $v = w - w' \in \mathcal{V}_0$. Orthogonality of ζ therefore makes $T_\zeta(w) = T_\zeta(w')$. Conversely, if $\zeta^\top v \neq 0$ for some $v \in \mathcal{V}_0$, every interior w admits a sufficiently small nonzero t such that $w + tv$ stays in the simplex interior. Its density is unchanged, but its target changes by $t\zeta^\top v$. This proves both directions.

For positive w, $f_j/p_w \leq 1/w_j$ on the density support. Each score component in (11) is therefore square integrable. For $a \in \mathbb{R}^{K-1}$, $a^\top I_w a = 0$ exactly when $\sum_{j<K} a_j(f_j - f_K) = 0$ almost everywhere. Its corresponding full null direction is $(a_1, \ldots, a_{K-1}, -\sum_{j<K} a_j)$. The target annihilates this null space exactly when β_ζ belongs to the range of I_w. When it does, $\phi_* = \beta_\zeta^\top I_w^+ U$ is centered and satisfies $\mathbb{E}_w(\phi_* U) = \beta_\zeta$. Any other centered square-integrable ϕ with the same covariance has $\phi - \phi_*$ orthogonal to the score span. Hence $\mathbb{E}_w(\phi^2) = \mathbb{E}_w(\phi_*^2) + \mathbb{E}_w\{(\phi - \phi_*)^2\}$, giving the bound. This is an influence-function variance bound, not a claim that a particular adaptive estimator attains it. Its sample-mean asymptotic variance factor is conditional on fixed kernels; nuisance estimation is not covered.

Penalized uniqueness and likelihood identification. For $\tau > 0$ and every $r_j > 0$, consider

$$L_n(w) = \sum_i \log\{\textstyle\sum_j w_j f_j(x_i)\} + \tau \sum_j r_j \log w_j.$$

Assume all kernel values at observed points are finite, and each point has positive density under some kernel. The prior term forces the objective to minus infinity at the boundary; its Hessian adds strictly positive curvature to the negative log likelihood. There is a unique interior penalized maximizer, even for duplicate kernels. This optimization fact does not alter the likelihood null space or establish branch-mass identification. An empirical SVD can diagnose numerical rank for chosen evaluation points, but it cannot prove population identification.

B Geometry, Coordinates, and a Temporal Extension

Proposition 5 (Coordinate transformation of branch targets)**.** *Let $\gamma_b : [0,1] \to \mathbb{R}^d$ be a one-to-one regular C^1 curve. For an increasing C^1 diffeomorphism $s = \varphi(u)$, put*

$$\widetilde{\gamma}_b(u) = \gamma_b\{\varphi(u)\}, \qquad \widetilde{h}_b(u) = h_b\{\varphi(u)\}\varphi'(u), \qquad \widetilde{K}_b(\cdot \mid u) = K_b\{\cdot \mid \varphi(u)\}.$$

Substitution in the branch integral leaves its probability law unchanged. For an exact branch, the chain rule gives

$$\frac{\widetilde{h}_b(u)}{\|\dot{\widetilde{\gamma}}_b(u)\|} = \frac{h_b\{\varphi(u)\}}{\|\dot{\gamma}_b\{\varphi(u)\}\|},$$

which is the same density per unit physical arc length. Normalized arc length is

$$u_b(s) = \frac{\int_0^s \|\dot{\gamma}_b(v)\|\, dv}{\int_0^1 \|\dot{\gamma}_b(v)\|\, dv}.$$

The distribution of $u_b(S)$ is invariant to the coordinate transport above when branch origin and orientation agree. Shared branch correspondence is still needed for a comparison across groups. In contrast, the transformed coordinate mean is $\int_0^1 \varphi^{-1}(s) h_b(s)\, ds$ and need not equal the original mean.

Proof. Substitute $s = \varphi(u)$ in the branch integral and coordinate mean. The chain rule gives $\|\dot{\widetilde{\gamma}}_b(u)\| = \|\dot{\gamma}_b\{\varphi(u)\}\|\varphi'(u)$, proving both the intrinsic density and normalized arc-length identities. □

For a full-rank counterexample, take $\gamma(s) = (s, 0)$, $h(s) = 1$ and $K(dx \mid s) = \mathcal{N}(dx; (s, 0), \sigma^2 I_2)$, with $\sigma > 0$. For $\kappa \neq 0$, put

$$\varphi_\kappa(u) = \frac{e^{\kappa u} - 1}{e^\kappa - 1}, \qquad \widetilde{h}_\kappa(u) = \frac{\kappa e^{\kappa u}}{e^\kappa - 1}, \qquad \widetilde{\gamma}_\kappa = \gamma \circ \varphi_\kappa.$$

Every κ gives the same observed law, but the transformed coordinate mean is $e^\kappa/(e^\kappa - 1) - 1/\kappa$, with limit $1/2$ and derivative $1/12$ at zero. Thus fixed endpoints and positive tube covariance alone do not identify a coordinate mean. On a self-overlapping immersion, the density relative to Hausdorff measure sums the Jacobian-adjusted contributions over coordinate preimages; Equation 5 uses the one-to-one case.

Skeleton and observed support. A finite collection of curve edges and declared vertices can be organized as a metric graph, allowing endpoints, junctions, and cycles. This describes the latent representation. A positive-mass full-rank Gaussian component has support throughout $\mathbb{R}^d$, so the topology of a noisy observed distribution differs from the connectivity of its centerlines. The formulation here uses declared structure and does not provide a topology-recovery theorem.

The tangent-normal covariance in Equation 4 is Euclidean. The locally adaptive normal distribution (LAND) instead uses a learned Riemannian metric, geodesic distance, and normalization relative to Riemannian volume (Arvanitidis et al., 2016). To use such a distribution as an SFD kernel, its law must be expressed relative to the same declared reference measure as the other components.

B.1 A Projected Fermat-Quintic Section

The Fermat quintic $\mathcal{Q} = \{\sum_{j=0}^4 z_j^5 = 0\} \subset \mathbb{CP}^4$ is a standard Calabi–Yau example (Hendi et al., 2024). Fixing projective coordinates $[-1 : z_1 : z_2 : 0 : 0]$ selects the affine curve $z_1^5 + z_2^5 = 1$, of two real dimensions. For $w = u + iv$, put

$$\begin{aligned} z_1 &= e^{2\pi i k/5}(\sin w)^{2/5}, \qquad z_2 = e^{2\pi i \ell/5}(\cos w)^{2/5}, \\ F(u, v; k, \ell) &= \big(\operatorname{Re} z_1, \operatorname{Re} z_2, (\operatorname{Im} z_1 + \operatorname{Im} z_2)/\sqrt{2}\big), \end{aligned} \tag{15}$$

where $k, \ell \in \{0, \ldots, 4\}$ and the powers use the principal complex logarithm. Since $\sin^2 w + \cos^2 w = 1$, every generated pair satisfies the section equation. Figure 3 plots F on $u \in [0.025, \pi/2 - 0.025]$, $v \in [-1, 1]$.

The observation law uses 13 equally spaced fixed values of v for each of the 25 root pairs. Each resulting curve has uniform u, mass $1/325$ and Gaussian covariance $0.035^2 I_3$. Thus an SFD with one-dimensional coordinate domains supplies a finite-curve approximation to the projected patches. Its full-rank observation law is defined in $\mathbb{R}^3$; its latent sampling rule is a coordinate-grid mixture rather than Calabi–Yau volume. The example concerns a projected section, not the full six-real-dimensional manifold or its Ricci-flat metric.

In Figure 3, colors identify the first root index k. The smooth patch display uses continuous v, while its observation law uses the finite grid described above.

B.2 A Temporal Extension

A possible temporal model would couple the observation kernels to a process on an oriented metric graph. An edge occupancy density $\rho_b(s, t)$ could satisfy

$$\partial_t \rho_b = -\partial_s j_b, \qquad j_b = v_b \rho_b - \partial_s(D_b \rho_b), \qquad D_b > 0.$$

Such a model also requires an initial law, probability-conserving vertex conditions, and any vertex-mass dynamics. Graph diffusion has established generator and vertex-boundary constructions (Kostrykin et al., 2012). Estimating drift, diffusion, or passage times would require time-ordered trajectories, sampling intervals, trajectory identifiers, and a treatment of sampling bias. Static occupancy alone does not identify those quantities. This is an extension of the observation model, not a fitted result in this paper.

C Figure Construction and Additional Examples

C.1 Construction of the Main-Text Displays

Figure 1 uses 32 grayscale contours selected inside an annotated foreground region, plus photo-guided paths for the outer pose, microphone and cable. Curve weights are proportional to arclength, occupancy is uniform in normalized arclength, and kernel covariance is $0.014^2 I_2$. The 18,000 draws use seed 2026102205. Its density display uses 4096 midpoint nodes per curve, a 500×500 grid and Gaussian convolution. Coordinates are scaled image-plane coordinates; photographic brightness does not set component mass.

Figure 2 uses the circular-tube law specified in the main text, with 16,000 draws and seed 2026102206. Its 64×64 bivariate histogram displays $[-2.25, 2.25]^2$, containing 15,853 observations; the remaining 147 draws are outside this window. Bin density divides by the full sample size and bin area. All radii enter the lower histogram. Figure 3 uses 18,000 draws and seed 2026102207 from the finite-curve law in Appendix B.1.

Figure 4 uses the first two coordinates of a ten-dimensional correlated Gaussian, a ten-dimensional skew mixture and a two-dimensional banana transform. All 500 stored training observations from repetition 1 of each reference scenario enter the histograms. The 24×24 joint bin densities use count divided by sample size and bin area and integrate to one. The lower panels are first-coordinate marginals.

C.2 Three-Dimensional and Other Specified Laws

Figure 10 shows a three-dimensional law, retaining its third-coordinate observations. Its localized Gaussian components have positive covariance, whereas an exact atom in Definition 1 is a point mass. The main-text density surfaces in Figure 5 use the first two coordinates instead. Their displayed heights are proportional to $(p/\max p)^a$, with panel-specific exponents $a = 0.28, 0.1664, 0.36, 0.36$ for the snowflake, face, maze and ECG-like laws. Colors also use panel-specific nonlinear relative scales. The face uses tube SD 0.070 and blob SD 0.105, different from the narrower sample illustration below. These displays are specified laws, without fitted-method comparisons.

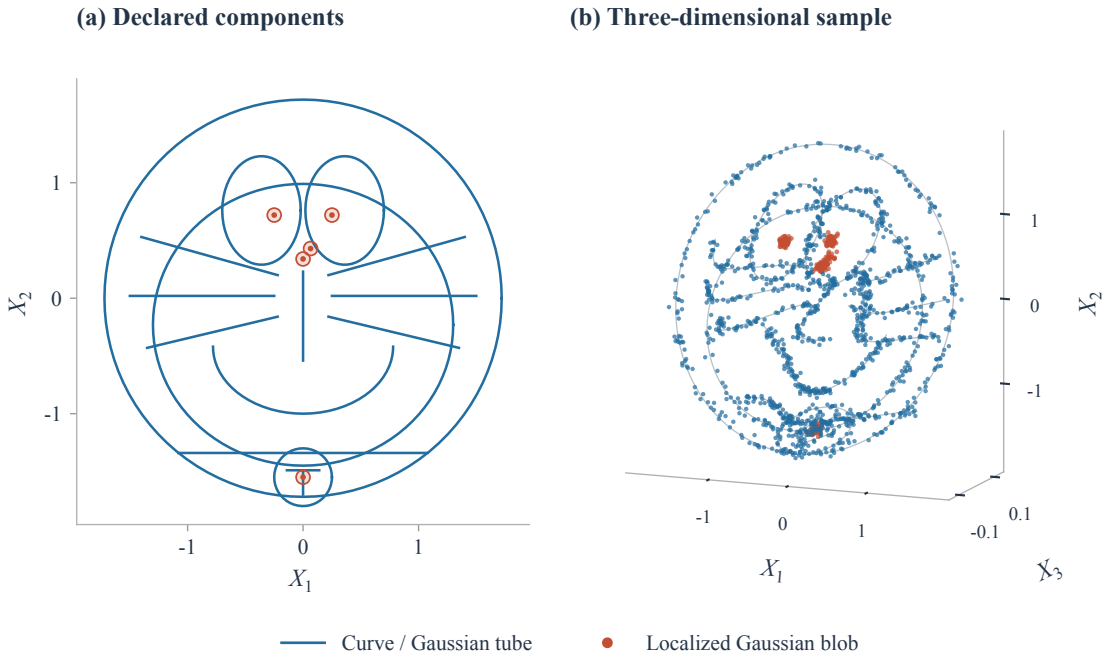


Figure 10: A stylized-face SFD with 16 Gaussian tubes and five localized Gaussian components. Left: projected centerlines and blob contours at squared Mahalanobis distance four. Right: all 1,600 observations in three dimensions, with gray centerlines as guides. Blue and red identify tubes and blobs. Branch coordinates are uniform on $[0, 1]$; SDs in the first two coordinates are 0.020 for tubes and 0.030 for blobs, and the third-coordinate SD is 0.012. The right panel uses actual X_3 values with unequal display scales.

D Distributional Calculus

All statements use the finite component sets and measurable kernels of Definition 1. Exact atoms are Dirac laws; regularized blobs can be represented among the continuous residual components. Component labels are latent and need not partition the observed space.

Proposition 6 (Normalization and latent construction)**.** *The SFD law is the marginal of the following construction: draw a component Z with probabilities π_c; given an atom, set $X = \xi_a$; given a branch, draw S with density h_b and then $X \sim K_b(\cdot \mid S)$; given a residual, draw X with density q_r. In particular, $P_\theta(\mathcal{X}) = 1$.*

Proof. Each branch integral is a measure: for disjoint measurable sets, monotone convergence passes their sum through the nonnegative kernel integral. It has total mass $\int h_b(s) K_b(\mathcal{X} \mid s)\, ds = 1$. The finite nonnegative mixture therefore has total mass $\sum_c \pi_c = 1$. Conditioning the stated latent construction on Z gives exactly its probability on every measurable set, proving equality of laws. □

Proposition 7 (Mixed-density evaluation)**.** *For fixed atom locations and fixed exact branch maps, a common reference is*

$$\nu = \lambda_d + \sum_a \delta_{\xi_a} + \sum_{b \in \mathcal{B}_{\text{ex}}} (\gamma_b)_{\#} ds,$$

where ds is restricted to the finite-volume coordinate domain. If P_c denotes each normalized component law, then

$$p_\theta = \sum_c \pi_c \frac{dP_c}{d\nu}.$$

For a continuous component with Lebesgue density g_c, its common-reference density is $g_c\rho$, where $\rho = d\lambda_d/d\nu$. For an injective exact branch, put $\nu_{\Gamma,b} = (\gamma_b)_{\#}ds$. Its contribution is

$$\pi_b h_b\{\gamma_b^{-1}(x)\}\frac{d\nu_{\Gamma,b}}{d\nu}(x).$$

Thus overlapping reference contributions must be converted before addition.

Proof. The branch reference measures are finite and their finite sum with Lebesgue measure is sigma-finite. A coordinate null set has zero probability under $h_b(s)ds$, so its pushforward probability is dominated by its pushforward reference. All components are therefore dominated by ν. Linearity of the Radon–Nikodym derivative gives the mixture formula; its chain rule gives the continuous and injective-branch expressions. In particular, a continuous density contributes zero mass at an exact atom, even if its Lebesgue density there is positive. □

For a regular immersion of intrinsic dimension m_b with finite preimages, the area formula gives, at almost every point of its image,

$$\frac{dP_b}{d\mathcal{H}^{m_b}}(x) = \sum_{s:\gamma_b(s)=x}\frac{h_b(s)}{J_b(s)}, \qquad J_b(s) = \sqrt{\det\{D\gamma_b(s)^\top D\gamma_b(s)\}} > 0.$$

Relative to the branch's own pushforward reference, the density is the ratio of this sum to $\sum_{s:\gamma_b(s)=x} 1/J_b(s)$, not the unweighted sum of coordinate densities. The common-reference conversion still applies.

D.1 Expectation, Moments and Covariance

For completeness, Proposition 1 follows first for nonnegative g by conditioning on Z, then S, and using Tonelli's theorem. For integrable signed g, apply the same argument to g^+ and g^- and subtract. This proves the full identity without a Lebesgue-only assumption.

In the Gaussian-tube case, whenever the second moments exist,

$$\mu = \mathbb{E}_\theta X = \sum_a \pi_a\xi_a + \sum_b \pi_b\int h_b(s)\gamma_b(s)\,ds + \sum_r \pi_r\mathbb{E}_{q_r}X,$$

$$\mathbb{E}_\theta(XX^\top) = \sum_a \pi_a\xi_a\xi_a^\top + \sum_b \pi_b\int h_b(s)\{\gamma_b(s)\gamma_b(s)^\top + \Sigma_b(s)\}\,ds + \sum_r \pi_r\mathbb{E}_{q_r}(XX^\top).$$

The covariance is $\mathbb{E}_\theta(XX^\top) - \mu\mu^\top$. A branch covariance therefore separates variation of $\gamma_b(S)$ from the average conditional noise covariance $\int h_b(s)\Sigma_b(s)\,ds$. If the component moment generating functions and the following integrals are finite near the origin,

$$M_X(t) = \sum_a \pi_a e^{t^\top\xi_a} + \sum_b \pi_b\int h_b(s)e^{t^\top\gamma_b(s)+t^\top\Sigma_b(s)t/2}\,ds + \sum_r \pi_r M_{q_r}(t).$$

Both formulas are choices of g in the master identity.

D.2 Marginal and Conditional Laws

Let J be a nonempty proper subset of coordinate indices, put $Y = X_{-J}$, and write T_J for projection onto X_J.

Proposition 8 (Marginal law)**.** *For arbitrary SFD kernels,*

$$(T_J)_{\#}P_\theta = \sum_a \pi_a\delta_{\xi_{a,J}} + \sum_b \pi_b\int h_b(s)(T_J)_{\#}K_b(\cdot\mid s)\,ds + \sum_r \pi_r(T_J)_{\#}Q_r.$$

For Gaussian tubes, the projected branch kernel is $\mathcal{N}(\gamma_{b,J}(s), \Sigma_{b,JJ}(s))$. Exact branches instead use their projected pushforward laws, which may change dimension or acquire atoms.

Proof. Evaluate the original law on $T_J^{-1}(A)$ for a measurable set A. Projection commutes with the finite sum and nonnegative kernel integral. The Gaussian assertion is the marginal formula for a multivariate normal. □

Proposition 9 (Gaussian branch conditional law)**.** *For a full-rank Gaussian branch kernel, conditional on $(Z = b, S = s, Y = y)$, X_J is Gaussian with mean and covariance*

$$m_{J|-J,b}(s; y) = \gamma_{b,J}(s) + \Sigma_{b,J,-J}(s)\Sigma_{b,-J,-J}(s)^{-1}\{y - \gamma_{b,-J}(s)\},$$
$$V_{J|-J,b}(s) = \Sigma_{b,JJ}(s) - \Sigma_{b,J,-J}(s)\Sigma_{b,-J,-J}(s)^{-1}\Sigma_{b,-J,J}(s).$$

Proof. Positive definiteness makes the conditioning covariance block invertible. Completing the square in the joint Gaussian density gives the displayed mean and Schur-complement covariance. □

Proposition 10 (Full conditional mixture)**.** *Choose a common projected reference ν_{-J} that dominates the projected component laws and, for each regularized branch, its projected slice laws. Let $g_{a,-J}$, $g_{b,-J}(y \mid s)$ and $g_{r,-J}$ be their densities relative to this same reference, and let $p_{\theta,-J}$ be the total density. For $P_{\theta,-J}$-almost every y with positive density, put*

$$\tau_a(y) = \frac{\pi_a g_{a,-J}(y)}{p_{\theta,-J}(y)}, \qquad \tau_b(s \mid y) = \frac{\pi_b h_b(s) g_{b,-J}(y \mid s)}{p_{\theta,-J}(y)}, \qquad \tau_r(y) = \frac{\pi_r g_{r,-J}(y)}{p_{\theta,-J}(y)}.$$

For regularized branches and exact atoms, the full conditional law is

$$\mathbb{P}(X_J \in A \mid Y = y) = \sum_a \tau_a(y)\mathbf{1}_A(\xi_{a,J}) + \sum_b \int \tau_b(s \mid y) K_{b,J|-J}(A \mid s, y)\, ds + \sum_r \tau_r(y) Q_{r,J|-J}(A \mid y).$$

For Gaussian tubes the branch conditional is given by Proposition 9. Exact branches whose slice laws lack common domination use a posterior coordinate measure obtained by disintegration, rather than the displayed density relative to ds.

Proof. Disintegrate the joint latent-observed law with respect to Y. The three Bayes numerators are its atom, branch-coordinate and residual contributions; their sum/integral is $p_{\theta,-J}$. They therefore normalize to one. Integrating each component's conditional law against these posterior weights gives the assertion. The exceptional set is projected-law null, as required for regular conditional distributions. □

For Gaussian slices under a mixed projected reference, $g_{b,-J}(y \mid s) = \phi_{d-|J|}\{y; \gamma_{b,-J}(s), \Sigma_{b,-J,-J}(s)\}\rho_{-J}(y)$, with $\rho_{-J} = d\lambda_{d-|J|}/d\nu_{-J}$. That conversion also applies to continuous residuals. Conditional simulation uses $\tau_b(s \mid y)$, whereas full-observation EM uses $\tau_b(s \mid x)$; they condition on different observations.

D.3 Special-Case Embeddings

Proposition 11 (Special-case status)**.** *The following restrictions give exact equalities of laws or estimators.*

1. *Atoms alone give finite discrete distributions. An atom and a specified continuous residual give the corresponding atom-inflated mixture.*
2. *Residual-only SFDs contain normal, product, heavy-tailed and finite Gaussian-mixture laws by choosing the corresponding normalized q_r.*
3. *A residual-only mixture of n fixed data-centered Gaussian kernels with weights $1/n$ is exactly that KDE estimator.*
4. *One Gaussian tube gives Proposition 2; specified multiple tubes give mixtures of those laws. A Gaussian factor-analyzer distribution enters as its integrated residual Gaussian density.*

5. *Exact kernels $\delta_{\gamma_b(s)}$ give coordinate-pushforward laws under an intrinsic or mixed reference, not ambient Lebesgue densities.*

6. *A specified copula density with its marginal densities can be chosen as a residual law. This does not identify SFD itself as a copula construction.*

A learned metric or temporal/path-functional model is an adjacent connection until its extra structure is explicitly included in a normalized kernel or dynamic extension.

Proof. Set inactive weights to zero and substitute the stated component laws in Definition 1. Each retained law is normalized. The KDE assertion holds conditional on its chosen centers and bandwidth; it is an estimator equality, not an additional parametric generating model. The pushforward case is interpreted using Proposition 7. Extra metric or temporal structure is not implied by a static mixture restriction. □

E Regular Likelihood Expansion and Contrast Tests

This section proves Theorem 1 for the whole estimated finite chart, rather than holding estimated noise or geometry fixed. The fixed component set and common support are part of its assumptions.

E.1 Conditions for Theorem 1

Let $X_1, \dots, X_n$ be iid from P_{θ_0} in a finite-dimensional SFD family $p_\theta = dP_\theta/d\nu$. In an open neighborhood of the interior parameter θ_0, assume:

(R1) The component set, atom supports, coordinate domains and effective support are fixed, with one common dominating measure ν.

(R2) The chart is identifiable, active weights are bounded away from zero, and active Gaussian tube covariance eigenvalues have positive lower and finite upper bounds.

(R3) The log density is C^3 almost everywhere on its support, its first three derivatives have integrable local uniform envelopes, and derivatives pass through branch integrals and integration against ν.

(R4) The score $U_0 = \nabla \log p_{\theta_0}$ has finite, positive-definite information $I_0 = \mathbb{E}_{\theta_0}(U_0 U_0^\top)$.

(R5) An interior local likelihood maximizer $\hat{\theta}_n$ exists in this chart with probability tending to one and is consistent for θ_0.

Consistency in (R5) is a separate input, not a consequence of local smoothness. Data-selected topology, zero weights, moving singular supports and covariance collapse require separate arguments. Transfer from a numerical likelihood also needs score-error control at the root-n scale (Appendix F).

E.2 Proof of the Regular Finite-SFD Likelihood Theorem

Write $P_n f = n^{-1} \sum_i f(X_i)$, $U_0(x) = \nabla \log p_{\theta_0}(x)$ and $H_0(x) = \nabla^2 \log p_{\theta_0}(x)$. Differentiating normalization twice, using the permitted integration interchanges, gives

$$\mathbb{E}_{\theta_0} U_0 = 0, \qquad \mathbb{E}_{\theta_0} H_0 = -\mathbb{E}_{\theta_0}(U_0 U_0^\top) = -I_0.$$

The finite information assumption gives the score central limit theorem, $\Delta_n = n^{-1/2} \sum_i U_0(X_i) \rightsquigarrow \mathcal{N}(0, I_0)$. For any fixed $M < \infty$, Taylor's theorem and the third-derivative envelope M_3 give, uniformly over $\|u\| \le M$,

$$\ell_n(\theta_0 + u/\sqrt{n}) - \ell_n(\theta_0) = u^\top \Delta_n + \tfrac{1}{2} u^\top (P_n H_0) u + R_n(u), \qquad \sup_{\|u\| \le M} |R_n(u)| \le C_M n^{-1/2} P_n M_3 = o_p(1).$$

Since $P_n H_0 \to_p -I_0$, this is the uniform local quadratic expansion

$$\ell_n(\theta_0 + u/\sqrt{n}) - \ell_n(\theta_0) = u^\top \Delta_n - \tfrac{1}{2} u^\top I_0 u + o_p(1). \tag{16}$$

Consistency and interiority give $\nabla \ell_n(\hat{\theta}_n) = 0$ in a neighborhood of θ_0 with probability tending to one. Use the segment-averaged Hessian, not a common scalar mean-value point:

$$\overline{H}_n = \frac{1}{n} \int_0^1 \nabla^2 \ell_n\{\theta_0 + t(\hat{\theta}_n - \theta_0)\}\, dt.$$

The fundamental theorem of calculus gives exactly

$$0 = \Delta_n + \overline{H}_n \sqrt{n}(\hat{\theta}_n - \theta_0).$$

The third-derivative envelope bounds $\|\overline{H}_n - P_n H_0\|$ by a constant times $\|\hat{\theta}_n - \theta_0\| P_n M_3 = o_p(1)$, so $\overline{H}_n \to_p -I_0$. Its inverse is bounded in probability. Inverting the score equation first establishes root-n consistency, then the linear representation and Gaussian limit in Equation 13. Differentiability of ψ gives

$$\sqrt{n}\{\psi(\hat{\theta}_n) - \psi(\theta_0)\} = \dot{\psi}(\theta_0) I_0^{-1} \Delta_n + o_p(1),$$

and hence the stated covariance. This proves the theorem.

E.3 Smooth Contrasts

For fixed component labels and coordinates, smooth targets include π_b, $\sum_{a \in A_0} \pi_a$, $\int s h_b(s)\, ds$ and $\int \mathrm{tr}\{\Sigma_{\perp,b}(s)\} h_b(s)\, ds$ when their parameter derivatives exist. Independent group-specific fits can be combined in one product chart; contrast variances then add the group contributions. Coordinate-dependent targets also use the conventions of Proposition 5.

Proposition 12 (Regular tests for SFD contrasts)**.** *Under Theorem 1, let $h : \Theta \to \mathbb{R}^q$ be C^2 near θ_0 with full row-rank $A = \dot{h}(\theta_0)$, and suppose $h(\theta_0) = 0$. Let a constrained local likelihood maximizer $\tilde{\theta}_n$ be consistent on this interior null manifold, and let $\hat{V}_n \to_p I_0^{-1}$. Then*

$$W_n = n h(\hat{\theta}_n)^\top \{\dot{h}(\hat{\theta}_n) \hat{V}_n \dot{h}(\hat{\theta}_n)^\top\}^{-1} h(\hat{\theta}_n) \;\rightsquigarrow\; \chi^2_q,$$
$$\Lambda_n = 2\{\ell_n(\hat{\theta}_n) - \ell_n(\tilde{\theta}_n)\} \;\rightsquigarrow\; \chi^2_q.$$

Proof. Put $B = A I_0^{-1} A^\top \succ 0$. The delta method gives $\sqrt{n} h(\hat{\theta}_n) \rightsquigarrow \mathcal{N}(0, B)$, and the estimated middle covariance tends to B. Its quadratic form proves the Wald result.

For the likelihood ratio, the C^2 full-rank null is locally a smooth coordinate chart by the implicit function theorem. The same consistent local-score argument as above, restricted to that chart, yields root-n consistency of the constrained estimator. Its limiting tangent space is $\mathcal{T} = \{u : Au = 0\}$. Let $z_n = I_0^{-1} \Delta_n$ and define its I_0-orthogonal projection

$$P_{\mathcal{T}} z = z - I_0^{-1} A^\top B^{-1} A z.$$

Completing the square in Equation 16 shows that its unrestricted and constrained maximizers are respectively z_n and $P_{\mathcal{T}} z_n$, up to $o_p(1)$. Root-n consistency allows the uniform compact expansion to be evaluated at both random maximizers. Twice their quadratic maximum difference is therefore

$$\Lambda_n = (A z_n)^\top B^{-1} (A z_n) + o_p(1).$$

Since $A z_n \rightsquigarrow \mathcal{N}(0, B)$, this tends to χ^2_q. □

The C^2 null-chart assumption is stated explicitly for the likelihood-ratio argument. A first-order differentiable contrast suffices for the Wald part. These calibrations concern regular likelihood estimation in the specified model; they do not calibrate zero weights, selected topology or singular covariance limits.

Misspecification and penalties. For a consistent interior estimator targeting a regular population optimum θ_* under a possibly misspecified law, an analogous centered-score expansion gives covariance $H^{-1} J H^{-\top}$, where $H = -\mathbb{E}\nabla^2 \log p_{\theta_*}(X)$ and $J = \mathbb{E}\{U_{\theta_*}(X) U_{\theta_*}(X)^\top\}$. This requires the corresponding smoothness, nonsingularity and central-limit assumptions at that target. A penalty with gradient $o_p(\sqrt{n})$ and Hessian $o_p(n)$ locally leaves the first-order limit unchanged. A non-negligible penalty instead changes the expansion or its centering. Neither a sandwich estimate nor prior curvature repairs likelihood nonidentification.

F Finite-Grid Optimization and Approximation

F.1 Proof of Fixed-Quadrature GEM Monotonicity

For Proposition 3, write $p_\theta(x, z)$ for the normalized finite-grid joint density and $r_i(z) = p_{\theta^{\text{old}}}(x_i, z)/p_{\theta^{\text{old}}}(x_i)$ for the old posterior. On labels with $r_i(z) > 0$, Jensen's inequality gives

$$\log p_\theta(x_i) \geq \sum_{z: r_i(z)>0} r_i(z) \log \frac{p_\theta(x_i, z)}{r_i(z)}.$$

The bound is exact at the old iterate. Summing and subtracting the fixed penalty gives $F_n(\theta) \geq Q(\theta \mid \theta^{\text{old}}) - \text{Pen}(\theta) + \sum_i \text{Ent}(r_i)$. The entropy is constant during the step. A nondecreasing penalized Q therefore implies a nondecreasing observed objective. Zero old posterior terms use $0 \log 0 = 0$; a proposed zero likelihood on a positive old posterior cannot improve a finite auxiliary objective. The branch-node term uses $\log \kappa_{bm} = \log w_{bm} + \log h_b(s_{bm}) - \log C_b$ throughout. Changing the grid or dropping its normalizer changes this argument's objective and is not covered by the proposition.

F.2 Numerical Likelihood and Root-n Inference

Proposition 13 (Transfer from a numerical likelihood)**.** *Under Theorem 1, let $\hat{\theta}_n$ be its consistent interior exact-likelihood maximizer, and let $\tilde{\theta}_n$ be consistent for the same target under a differentiable numerical criterion $\tilde{\ell}_n$. On a fixed neighborhood N of θ_0, assume*

$$\|\nabla \tilde{\ell}_n(\tilde{\theta}_n)\| = o_p(\sqrt{n}), \qquad \sup_{\theta \in N} \|\nabla \tilde{\ell}_n(\theta) - \nabla \ell_n(\theta)\| = o_p(\sqrt{n}).$$

Then $\sqrt{n}(\tilde{\theta}_n - \hat{\theta}_n) = o_p(1)$. If also

$$\sup_{\theta \in N} n^{-1} \|\nabla^2 \tilde{\ell}_n(\theta) - \nabla^2 \ell_n(\theta)\| = o_p(1),$$

then $-n^{-1} \nabla^2 \tilde{\ell}_n(\tilde{\theta}_n) \to_p I_0$.

Proof. The two score conditions give $\nabla \ell_n(\tilde{\theta}_n) = o_p(\sqrt{n})$. Subtract the zero exact score at $\hat{\theta}_n$ and integrate the exact Hessian along their joining segment. Consistency and the local envelope argument in Appendix E make that Hessian divided by n tend to $-I_0$. Inverting the score difference proves the estimator claim. The additional Hessian discrepancy gives the information claim. □

The rates matter even for normalized densities. In the residual-only normal submodel, take $p_\theta(x) = \phi_1(x; \theta, 1)$ and $\tilde{p}_{n,\theta}(x) = \phi_1(x; \theta + \epsilon_n, 1)$ with $\epsilon_n = n^{-1/4}$. Under iid $\mathcal{N}(0, 1)$ data, the average criterion difference vanishes uniformly on compact parameter sets, average scores differ by $-\epsilon_n \to 0$, and Hessians agree. Yet $\tilde{\theta}_n - \hat{\theta}_n = -\epsilon_n$ and its root-n multiple diverges. Transfer to the continuous law requires average score error $o_p(n^{-1/2})$, not just $o_p(1)$. A fixed normalized grid can instead be treated as the statistical model in its own right.

F.3 Mixture and Responsibility Perturbations

Proposition 14 (Mixture stability on a fixed stratum set)**.** *Let $Q = \sum_c \pi_c Q_c$ and $\tilde{Q} = \sum_c \tilde{\pi}_c \tilde{Q}_c$ have the same finite component index set. For $d_{\text{TV}}(P, Q) = \sup_A |P(A) - Q(A)|$,*

$$d_{\text{TV}}(Q, \tilde{Q}) \leq \tfrac{1}{2} \sum_c |\pi_c - \tilde{\pi}_c| + \sum_c (\pi_c \wedge \tilde{\pi}_c) d_{\text{TV}}(Q_c, \tilde{Q}_c).$$

If all components are supported on a set of diameter $D < \infty$, then for $p \geq 1$,

$$W_p^p(Q, \tilde{Q}) \leq D^p \tfrac{1}{2} \sum_c |\pi_c - \tilde{\pi}_c| + \sum_c (\pi_c \wedge \tilde{\pi}_c) W_p^p(Q_c, \tilde{Q}_c).$$

Proof. Set $\alpha_c = \pi_c \wedge \tilde{\pi}_c$. Couple the shared mass α_c within component c using a total-variation-optimal coupling. Couple the remaining mass arbitrarily. Its total weight is $1 - \sum_c \alpha_c = \frac{1}{2}\sum_c |\pi_c - \tilde{\pi}_c|$, giving the first bound. For Wasserstein distance, use optimal or arbitrarily close-to-optimal component couplings and transport the unmatched mass at cost at most D^p per unit mass. Taking the infimum gives the second bound. The diameter assumption is not asserted for unbounded Gaussian tubes. □

Proposition 15 (Relative quadrature error controls log likelihood)**.** *Suppose $p_\theta(x_i)$ and $\tilde{p}_\theta(x_i)$ are finite and positive and, uniformly over $\theta \in \Theta_0$ and all observations,*

$$|\tilde{p}_\theta(x_i)/p_\theta(x_i) - 1| \le \epsilon < 1.$$

Then

$$\sup_{\theta\in\Theta_0} |\tilde{\ell}_n(\theta) - \ell_n(\theta)| \le -n\log(1-\epsilon) \le 2n\epsilon \quad \text{if } \epsilon \le 1/2.$$

Proof. Each density ratio lies between $1-\epsilon$ and $1+\epsilon$. Take logarithms, sum, and use $\log(1+\epsilon) \le -\log(1-\epsilon)$. The last bound follows from $-\log(1-\epsilon) \le 2\epsilon$ on $[0, 1/2]$. This is a criterion bound; it does not establish the score rates in Proposition 13. □

Proposition 16 (Softmax responsibility perturbation)**.** *For finite log-score vectors $a, \tilde{a}$, let $r_z = e^{a_z}/\sum_u e^{a_u}$ and $\tilde{r}_z = e^{\tilde{a}_z}/\sum_u e^{\tilde{a}_u}$. If $\max_z |a_z - \tilde{a}_z| \le \delta$, then*

$$e^{-2\delta} \le \tilde{r}_z/r_z \le e^{2\delta}, \qquad \sum_z |\tilde{r}_z - r_z| \le e^{2\delta} - 1.$$

Proof. Exponentiation bounds each numerator ratio by $e^{\pm\delta}$; summation gives the same bounds for the normalizing-constant ratio. Division gives $e^{\pm 2\delta}$. Thus $|\tilde{r}_z - r_z| \le (e^{2\delta} - 1)r_z$; summing proves the result. □

G Tube-Distance Concentration

This result isolates the probability of leaving a declared tube from the geometric distance estimate used to define that tube. In particular, a positive reach condition alone is not a derivation of a residual bound.

Proposition 17 (Tube-distance concentration from residual tails)**.** *Let $\Gamma_b = \gamma_b(\mathcal{S}_b)$ and $d_{\Gamma_b}(x) = \inf_{y\in\Gamma_b} \|x - y\|$. On a measurable coordinate subset U_b of positive branch-coordinate probability, suppose the observation is $X = \gamma_b(S) + t_b(S)\varepsilon_{\parallel,b} + N_b(S)\varepsilon_{\perp,b}$. Let $0 < M_{N,b} = \sup_{s\in U_b} \|N_b(s)\|_{\mathrm{op}} < \infty$. Assume a nonnegative measurable remainder R_b and radius $\rho_b > 0$ satisfy*

$$d_{\Gamma_b}(X) \le M_{N,b}\|\varepsilon_{\perp,b}\| + R_b \quad \text{whenever } S \in U_b \text{ and } M_{N,b}\|\varepsilon_{\perp,b}\| + R_b < \rho_b.$$

For $0 < r < \rho_b$ and $0 < \eta < 1$, set $a = (1-\eta)r/M_{N,b}$ and

$$\begin{aligned}\Delta_b(r,\eta) = \mathbb{P}(S \notin U_b \mid Z = b) \\ + \operatorname*{ess\,sup}_{s\in U_b} \mathbb{P}(\|\varepsilon_{\perp,b}\| > a \mid S = s, Z = b) \\ + \operatorname*{ess\,sup}_{s\in U_b} \mathbb{P}(R_b > \eta r \mid S = s, Z = b).\end{aligned}$$

Essential suprema use the branch-coordinate law. Then

$$\mathbb{P}\{d_{\Gamma_b}(X) > r \mid Z = b\} \le \Delta_b(r,\eta).$$

For n_b observations conditional on branch b, the probability that at least one leaves the radius-r tube is at most $n_b\Delta_b(r,\eta)$. For n draws from the mixture and an active branch subset $\mathcal{B}_0$,

$$\mathbb{P}\{\exists i : Z_i \in \mathcal{B}_0, d_{\Gamma_{Z_i}}(X_i) > r\} \le n\sum_{b\in\mathcal{B}_0} \pi_b\Delta_b(r,\eta).$$

Proof. If $S \in U_b$, $\|\varepsilon_{\perp,b}\| \le a$ and $R_b \le \eta r$, the assumed bound is at most $r < \rho_b$. The escape event is therefore contained in the union of the three complement events in Δ_b. Conditioning on S and applying the union bound proves the first assertion. Union bounds over observations and then mixture labels give the other two. They do not require independence beyond the stated one-observation laws. □

Corollary 18 (Gaussian residual-tail specialization)**.** *In Proposition 17, suppose conditionally on $S = s$ that $\varepsilon_{\perp,b} = A_b(s)G$ with $G \sim \mathcal{N}(0, I_{r_b^\varepsilon})$. Set*

$$\bar{m}_b = \operatorname*{ess\,sup}_{s \in U_b} \mathbb{E}\|A_b(s)G\|, \qquad \bar{\sigma}_b = \operatorname*{ess\,sup}_{s \in U_b} \|A_b(s)\|_{\mathrm{op}}.$$

If $0 < \bar{\sigma}_b < \infty$, the normal-residual term in Δ_b is bounded by

$$\exp\left\{-\frac{(a - \bar{m}_b)_+^2}{2\bar{\sigma}_b^2}\right\}.$$

If $\bar{\sigma}_b = 0$, that term is zero for $a > 0$.

Proof. For fixed s, the Gaussian function $g \mapsto \|A_b(s)g\|$ is $\|A_b(s)\|_{\mathrm{op}}$-Lipschitz. Gaussian Lipschitz concentration (Boucheron et al., 2013) bounds its upper tail above its mean by $\exp\{-t^2/(2\|A_b(s)\|_{\mathrm{op}}^2)\}$ for $t \ge 0$. Use the trivial bound one when a is below the mean, otherwise take $t = a - \mathbb{E}\|A_b(s)G\|$. Uniform bounds on the mean and Lipschitz constant give the display. Zero operator norm gives zero residual almost surely. □

Other observation kernels enter the same first proposition through their own residual tails. It is the kernel law and the assumed geometric remainder, not the name of the mixture family, that determine concentration.

H Experiment Details

The following settings define the four studies in Section 6. Numerical summaries use all recorded repetitions and retain the convergence status of every selected fit. The supplementary tables report the full prespecified density and marker controls.

Known-skeleton weight diagnostic. The five generating laws and component families are fixed before fitting. Each of 25 repetitions uses 1000 training and 1000 independent test observations. EM estimates component weights with a 64-node tube grid, a maximum of 300 iterations, maximum-weight-change tolerance 10^{-6} and minimum weight 10^{-12}. Gaussian, KDE and GMM controls share the observations; GMM sizes are $2, 4, 8, 16, 32$, with maximum 80 iterations, tolerance 10^{-5} and covariance regularization 10^{-5}. The base seed is 20260623, with repetition offset 10000 and zero-based scenario offset 101. The archived per-run and summary tables retain every method, convergence status and recovery result. This experiment does not estimate the skeleton, tube width or latent-coordinate error.

Two-arm study. The location μ_b^{loc} is a truncated-normal parameter, not the coordinate mean. The mean-position contrast changes it from 0.5 to 0.7 at scale 0.16 on $[0, 1]$, giving a generating mean contrast of about 0.189. Longitudinal convolution with tangent Gaussian noise has a closed form. The direct optimizer is L-BFGS-B with maxiter 300, ftol 10^{-10}, gtol 10^{-6}, maxls 40, mass-logit bounds $[-5, 5]$, location bounds $[0.1, 0.9]$, and width bounds $[0.025, 0.3]$. Every method and repetition is retained below; the oracle row is a generating-density reference, not a fitted competitor. The archived score rows also retain the observable $\mathbb{P}(X_2 > 0)$ comparison and all convergence flags, including GMM iteration limits.

Geometry and noise study. The learned directions are bounded to $[5, 75]$ and $[-75, -5]$ degrees. Three starts use angle pairs ± 20, ± 35, ± 50 degrees, with the recorded location and width starts. L-BFGS-B uses maxiter 500 and the same tolerances. The best finite training objective selects the fit. GMM uses three separately seeded starts, covariance regularization 10^{-5}, maxiter 2000 and tol 10^{-5}. KDE chooses 0.5, 1, or 2 times Scott's bandwidth on a deterministic 80/20 inner split, then refits on all training rows. The outer test set is never used for selection.

PBMC prediction diagnostic. The analysis uses the public 10x PBMC3k filtered count matrix. Filtering keeps genes detected in at least three training cells, and selects the 1000 largest training log-CP10k variances, breaking ties by gene index. Training means and SDs scale the data, values are clipped to $[-10, 10]$, and randomized PCA has two components. All 18 predeclared marker genes are excluded from predictor normalization and selection. Marker-group means use the same nonmarker library size. A cell is positive when the B-marker mean log-CP10k is at least 0.5 and strictly exceeds each competing marker-group mean. This fixed rule precedes the repeated analysis.

The MST model uses 24 landmarks and 12 quadrature nodes with the archived width/residual heuristic. Its small-mass-change stopping rule is not a joint objective certificate. All density controls and readouts share the outer train/test representation. Logistic predictors select inverse L2 penalty from $\{0.01, 0.1, 1, 10\}$ on the same stratified 75/25 inner training split, then refit all outer training cells. Dropping the eighth GMM posterior coordinate removes exact simplex redundancy. The two-coordinate density and marker readouts concern cell-level prediction within one collection; independent-donor and externally annotated validation are separate tasks.

I Parameter Intervals for the Saved Fits

The post-hoc interval diagnostic reuses all 900 saved SFD fits from the geometry/noise study: 600 use six parameters and 300 include the two estimated angles. The chart contains two mass logits relative to the residual, two coordinate-normal locations, two log transverse SDs, and the two angles when active. The origin, coordinate/tangent scales, residual law, and two-arm family remain supplied. No fit is repeated, and no test observations enter the interval calculation.

Let $H = \nabla^2_\theta\{-\ell_n(\theta)/n\}\big|_{\theta=\hat\theta}$ and $U_i = \nabla_\theta \log p_\theta(X_i)|_{\theta=\hat\theta}$. With $J = n^{-1}\sum_i U_i U_i^\top$, the two covariance estimates are

$$\widehat{V}_{\text{info}} = H^{-1}/n, \qquad \widehat{V}_{\text{sand}} = H^{-1}JH^{-\top}/n.$$

The inverse uses the complete fitted chart; estimated angles are not held fixed. For a target ψ, the delta-method variance is $\dot\psi(\hat\theta)\widehat{V}\dot\psi(\hat\theta)^\top$. The mass gradient uses the softmax row. The truncated-coordinate-mean derivative with respect to its normal location is $\mathrm{Var}(S \mid Z = +)/0.16^2$; the SD derivative with respect to log SD is σ_+. Contrasts add the independent alternative and reference variances. Intervals are untransformed normal Wald intervals without clipping. The three contrasts within a repetition share the reference fit.

Numerical checks. The original 300 training arrays regenerate identically. Source-version, saved-objective and target checks precede inference. Central differences of the average-loss gradient use relative steps 10^{-4}, 5×10^{-5}, and 2.5×10^{-5}. Relative Hessian changes and asymmetry must be at most 5×10^{-5}; the information eigenvalues must exceed 10^{-8} and the condition number must be below 10^8 in the stated chart. Fit and perturbed parameters remain interior, and maximum average-score magnitude must be at most 10^{-4}. No ridge or pseudoinverse repairs a failed matrix. All 900 saved fits pass these gates.

An independent audit constructs scalar-likelihood Hessians using relative steps 4×10^{-4} and 2×10^{-4} with Richardson cancellation. It uses numerical individual log-density derivatives and target derivatives at step 10^{-5}, rather than the analytic scores or production target Jacobian. All group-target standard errors agree with the saved-fit calculation under mixed relative/absolute tolerance 5×10^{-5} and 10^{-9}. Coverage, contrast variances and every summary cell are independently recounted. The complete record retains 9450 group/contrast intervals and 378 summary cells, each with 25 repetitions and zero invalid fits. Intervals from the same fit or shared reference are not pooled as independent trials.

Interpretation. Coverage refers to the generating targets, including under rotated geometry or Student-t_5 noise. A sandwich covariance estimates variation around a pseudo-true target under the corresponding regularity assumptions; it does not remove its discrepancy from the generating parameter. Figure 11 and Table 1 show every intended contrast/target pair. The accompanying summary data also retain all group parameters and cross-effects. With 25 repetitions, coverage is imprecisely estimated; even a count of 25/25 does not establish nominal calibration. This diagnostic accounts for two estimated directions, not topology selection or a general learned skeleton.

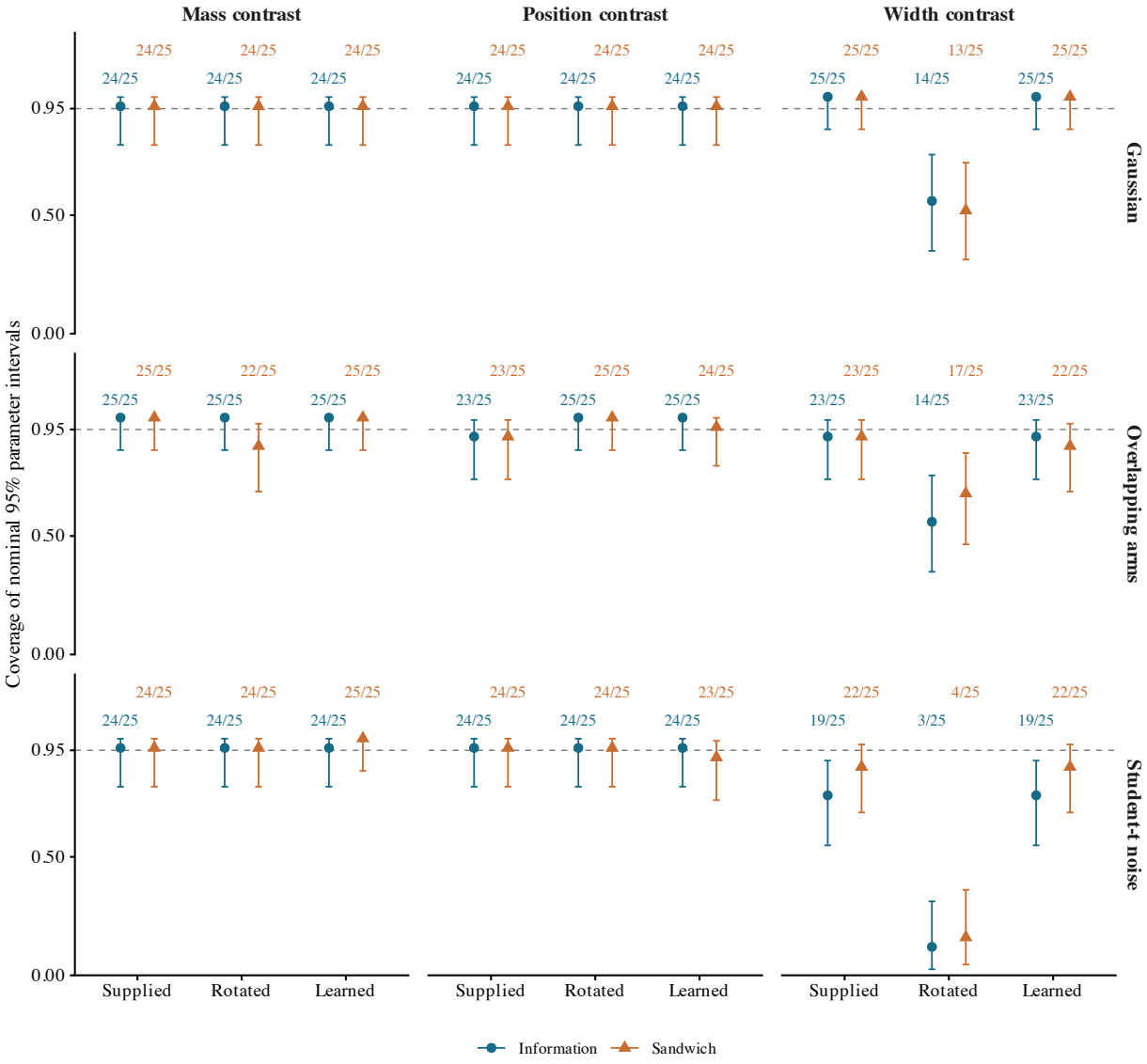


Figure 11: Post-hoc coverage of nominal 95% parameter intervals for the geometry/noise study. Columns show the three intended mass, coordinate-mean and transverse-SD contrasts; rows show Gaussian noise, overlapping arms, and equal-SD Student-t_5 noise. Points and counts use all 25 repetitions. Bars are exact binomial 95% Monte Carlo intervals for coverage, not parameter intervals; the dashed line marks 0.95. Both covariance choices use the same saved fit and complete six/eight-parameter chart. The 54 displayed cells follow the prespecified contrast/target pairs; the full numerical record retains all 378 cells.

Table 1: All intended contrast/target pairs in the saved geometry fits. Info. and Sand. give covered intervals out of 25 for information and sandwich covariance. Bias is the mean estimate minus the generating contrast; it is the same for both covariance choices. SE columns are mean parameter standard errors, not Monte Carlo SEs. Position is the truncated coordinate mean; SD is transverse spread. Cross-effects remain in the accompanying complete data.

Condition	Directions	Target	Info.	Sand.	Bias	SE (info.)	SE (sand.)
Gaussian	Supplied	Mass	24/25	24/25	0.00752	0.02354	0.02354
Gaussian	Supplied	Position	24/25	24/25	0.00173	0.01186	0.01173
Gaussian	Supplied	SD	25/25	25/25	0.00153	0.00733	0.00730
Gaussian	Rotated	Mass	24/25	24/25	0.00830	0.02389	0.02387
Gaussian	Rotated	Position	24/25	24/25	-0.00048	0.01206	0.01178
Gaussian	Rotated	SD	14/25	13/25	-0.01342	0.00751	0.00716
Gaussian	Learned	Mass	24/25	24/25	0.00756	0.02354	0.02355
Gaussian	Learned	Position	24/25	24/25	0.00171	0.01186	0.01174
Gaussian	Learned	SD	25/25	25/25	0.00154	0.00738	0.00738
Overlap	Supplied	Mass	25/25	25/25	0.00697	0.02596	0.02601
Overlap	Supplied	Position	23/25	23/25	-0.00149	0.01268	0.01268
Overlap	Supplied	SD	23/25	23/25	0.00305	0.00906	0.00903
Overlap	Rotated	Mass	25/25	22/25	0.01863	0.02954	0.02859
Overlap	Rotated	Position	25/25	25/25	0.00185	0.01549	0.01505
Overlap	Rotated	SD	14/25	17/25	-0.01498	0.01001	0.01069
Overlap	Learned	Mass	25/25	25/25	0.00497	0.02733	0.02744
Overlap	Learned	Position	25/25	24/25	-0.00135	0.01280	0.01281
Overlap	Learned	SD	23/25	22/25	0.00385	0.01191	0.01211
Student-t_5	Supplied	Mass	24/25	24/25	-0.00972	0.02373	0.02386
Student-t_5	Supplied	Position	24/25	24/25	0.00281	0.01191	0.01183
Student-t_5	Supplied	SD	19/25	22/25	-0.00655	0.00713	0.00819
Student-t_5	Rotated	Mass	24/25	24/25	-0.00688	0.02396	0.02396
Student-t_5	Rotated	Position	24/25	24/25	0.00064	0.01203	0.01186
Student-t_5	Rotated	SD	3/25	4/25	-0.02203	0.00722	0.00760
Student-t_5	Learned	Mass	24/25	25/25	-0.00979	0.02374	0.02387
Student-t_5	Learned	Position	24/25	23/25	0.00283	0.01191	0.01183
Student-t_5	Learned	SD	19/25	22/25	-0.00645	0.00718	0.00830

Table 2: Complete fixed-direction joint-fit interval diagnostic. Covered counts use 25 repetitions per row. MC intervals are exact binomial 95% intervals for coverage; empirical SD and mean SE concern the parameter or contrast estimator. Contrast rows subtract the independently sampled reference group. None of these rows describes learned-direction uncertainty.

Scope	Group	Target	Covered	95% MC interval	Empirical SD	Mean SE
Group	Reference	Mass	24/25	[0.796, 0.999]	0.01738	0.01670
Group	Reference	Position	24/25	[0.796, 0.999]	0.00823	0.00870
Group	Reference	Width	22/25	[0.688, 0.975]	0.00369	0.00309
Group	Mass	Mass	22/25	[0.688, 0.975]	0.02092	0.01673
Group	Mass	Position	25/25	[0.863, 1.000]	0.00486	0.00697
Group	Mass	Width	24/25	[0.796, 0.999]	0.00189	0.00240
Group	Position	Mass	24/25	[0.796, 0.999]	0.01324	0.01637
Group	Position	Position	24/25	[0.796, 0.999]	0.00794	0.00810
Group	Position	Width	25/25	[0.863, 1.000]	0.00285	0.00298
Group	Width	Mass	24/25	[0.796, 0.999]	0.01758	0.01798
Group	Width	Position	25/25	[0.863, 1.000]	0.00972	0.00932
Group	Width	Width	24/25	[0.796, 0.999]	0.00713	0.00669
Contrast	Mass	Mass	22/25	[0.688, 0.975]	0.02711	0.02364
Contrast	Mass	Position	25/25	[0.863, 1.000]	0.00884	0.01115
Contrast	Mass	Width	24/25	[0.796, 0.999]	0.00380	0.00392
Contrast	Position	Mass	24/25	[0.796, 0.999]	0.02030	0.02338
Contrast	Position	Position	23/25	[0.740, 0.990]	0.01095	0.01189
Contrast	Position	Width	23/25	[0.740, 0.990]	0.00515	0.00430
Contrast	Width	Mass	24/25	[0.796, 0.999]	0.02332	0.02454
Contrast	Width	Position	23/25	[0.740, 0.990]	0.01417	0.01275
Contrast	Width	Width	24/25	[0.796, 0.999]	0.00826	0.00737

Fixed directions in the original mechanism study. A separate post-hoc calculation uses all 100 joint fits from the earlier mechanism study. Its complete six-parameter covariance gives 300 group-target and 225 contrast-target intervals with no invalid fits. Table 2 retains all 21 summary cells. These are different saved fits from those in the geometry/noise study; their repetitions are not pooled.

J All Prespecified Density and Marker Controls

Every cell below is mean (Monte Carlo SE) over 25 runs. Density scores are in nats per observation; higher is better. Marker log loss is lower-is-better. PBMC differences are paired within split, and their SEs describe split variability in this one collection. No independent-subject uncertainty is estimated. Simulation score tables retain the same observations per method; the supplied structural information differs between the SFD ablations and generic density controls as described in the main text.

J.1 Fixed-geometry density scores

Method	Reference	Mass	Position	Width
Oracle	0.146 (0.007)	0.232 (0.008)	0.144 (0.007)	−0.079 (0.008)
SFD joint	0.143 (0.007)	0.229 (0.008)	0.142 (0.007)	−0.083 (0.008)
SFD fixed kernels	0.145 (0.007)	0.231 (0.008)	−0.095 (0.008)	−0.179 (0.007)
SFD oracle kernels	0.145 (0.007)	0.231 (0.008)	0.143 (0.007)	−0.080 (0.008)
Gaussian	−0.475 (0.007)	−0.451 (0.008)	−0.601 (0.009)	−0.528 (0.009)
GMM-4	0.132 (0.007)	0.218 (0.008)	0.129 (0.007)	−0.095 (0.008)
GMM-8	0.111 (0.007)	0.194 (0.008)	0.109 (0.008)	−0.117 (0.009)
KDE	−0.078 (0.009)	0.007 (0.008)	−0.114 (0.009)	−0.240 (0.010)

J.2 Geometry/noise density scores

J.2.1 Gaussian noise.

Method	Reference	Mass	Position	Width
SFD supplied	0.123 (0.006)	0.226 (0.007)	0.142 (0.010)	−0.088 (0.007)
SFD rotated	0.010 (0.005)	0.109 (0.008)	−0.020 (0.008)	−0.152 (0.007)
SFD learned angles	0.122 (0.006)	0.225 (0.007)	0.141 (0.010)	−0.089 (0.007)
Gaussian	−0.496 (0.005)	−0.454 (0.011)	−0.602 (0.008)	−0.537 (0.009)
GMM-4	0.112 (0.006)	0.216 (0.008)	0.128 (0.010)	−0.101 (0.007)
GMM-8	0.088 (0.007)	0.193 (0.008)	0.105 (0.012)	−0.127 (0.008)
KDE	−0.111 (0.016)	−0.001 (0.019)	−0.165 (0.020)	−0.263 (0.011)

J.2.2 Overlapping arms.

Method	Reference	Mass	Position	Width
SFD supplied	0.199 (0.008)	0.289 (0.011)	0.166 (0.005)	0.067 (0.005)
SFD rotated	0.139 (0.008)	0.212 (0.010)	0.045 (0.006)	0.043 (0.005)
SFD learned angles	0.198 (0.008)	0.288 (0.011)	0.165 (0.005)	0.065 (0.005)
Gaussian	−0.326 (0.011)	−0.307 (0.013)	−0.401 (0.011)	−0.360 (0.008)
GMM-4	0.185 (0.009)	0.277 (0.011)	0.153 (0.006)	0.055 (0.005)
GMM-8	0.165 (0.008)	0.253 (0.011)	0.132 (0.006)	0.030 (0.006)
KDE	0.009 (0.013)	0.085 (0.017)	−0.023 (0.011)	−0.074 (0.008)

J.2.3 Student-t noise.

Method	Reference	Mass	Position	Width
SFD supplied	0.157 (0.007)	0.264 (0.008)	0.173 (0.008)	−0.043 (0.007)
SFD rotated	0.023 (0.006)	0.130 (0.008)	−0.005 (0.007)	−0.121 (0.007)
SFD learned angles	0.157 (0.007)	0.263 (0.008)	0.173 (0.008)	−0.044 (0.007)
Gaussian	−0.510 (0.008)	−0.450 (0.009)	−0.611 (0.007)	−0.522 (0.007)
GMM-4	0.148 (0.007)	0.257 (0.009)	0.160 (0.008)	−0.052 (0.007)
GMM-8	0.129 (0.007)	0.236 (0.009)	0.138 (0.008)	−0.075 (0.007)
KDE	−0.184 (0.040)	−0.035 (0.037)	−0.171 (0.026)	−0.224 (0.007)

J.3 PBMC density and marker controls

Method	Test log density
Gaussian	−6.422 (0.006)
GMM-2	−5.649 (0.011)
GMM-4	−5.415 (0.010)
GMM-8	−5.367 (0.009)
KDE	−5.421 (0.009)
MST SFD	−5.770 (0.018)

Readout	Log loss	Change vs density
PCs	0.2526 (0.0036)	0.0017 (0.0006)
PCs + density	0.2509 (0.0033)	0.0000 (0.0000)
PCs + density + entropy	0.2257 (0.0041)	−0.0252 (0.0032)
PCs + density + SFD	0.2029 (0.0056)	−0.0481 (0.0044)
PCs + density + GMM memberships	0.1544 (0.0024)	−0.0965 (0.0030)

K External-Family Density Controls

These two archived comparisons characterize generic density estimators on classical and copula distributions. Each uses five repetitions with 1000 training and 1000 independent test observations per scenario. Oracle rows evaluate the known generating density; neither comparison fits an SFD. The full score and summary tables accompany the compilation sources. Finite scores from runs with iteration-limit warnings remain in the means; the archived status columns identify those runs.

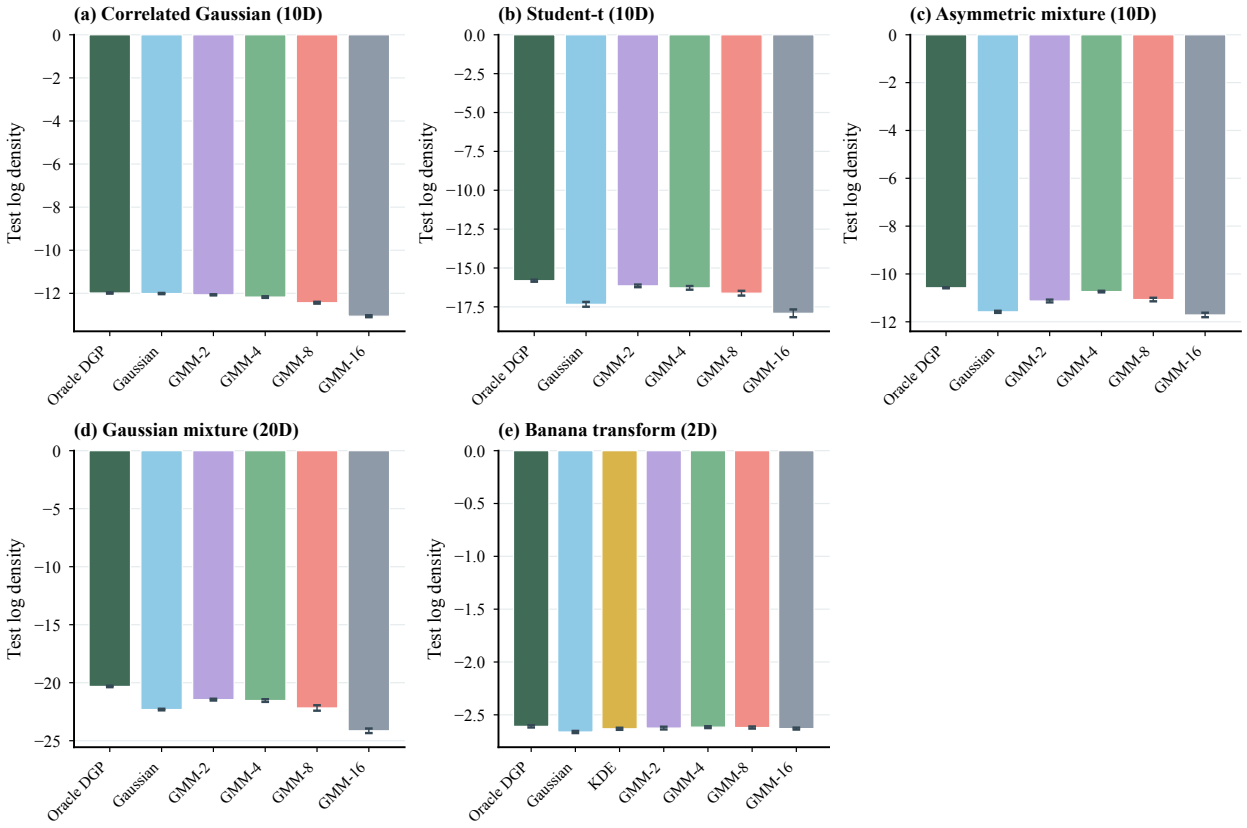


Figure 12: Classical distributions: correlated Gaussian, Student-t, asymmetric Gaussian mixture, Gaussian mixture and banana transform. Bars show mean test log density and one Monte Carlo SE over five repetitions. The oracle is the generating density; no SFD is fitted.

Classical-family settings. The first three distributions have ten coordinates, the Gaussian mixture has twenty, and the banana transform has two. GMM sizes are $2, 4, 8, 16$, with maximum 50 iterations, tolerance 10^{-5} and covariance regularization 10^{-5}. KDE is included only for the two-dimensional scenario. The base seed is 20260624. These reduced-scale controls are separate from the 25-repetition known-skeleton diagnostic.

Copula-family settings. The Gaussian and Student-t copulas have ten coordinates; Clayton has five. The Gaussian correlation is 0.55. The Student-t copula has correlation 0.45 and 4.5 degrees of freedom. The Clayton parameter is 1.6. GMM sizes are $2, 4, 8, 16$, with maximum 60 iterations, tolerance 10^{-5} and covariance regularization 10^{-5}; KDE is included for Clayton only. The base seed is 20260624. These controls do not include a fitted vine copula.

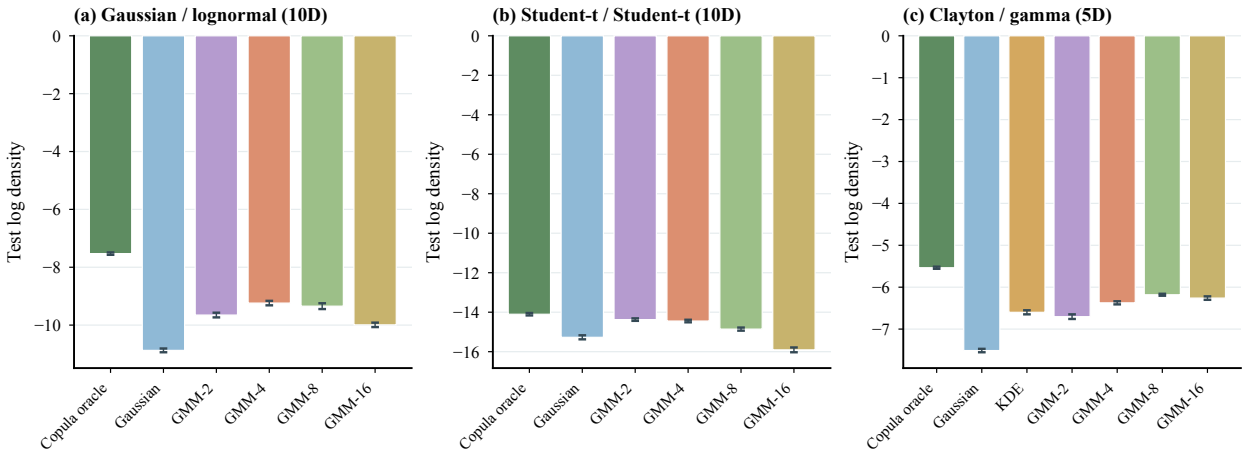


Figure 13: Copula families: Gaussian with lognormal margins, Student-t with Student-t margins, and Clayton with gamma margins. Bars show mean test log density and one Monte Carlo SE over five repetitions. The oracle includes the copula and marginal densities; no SFD is fitted.